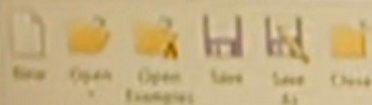
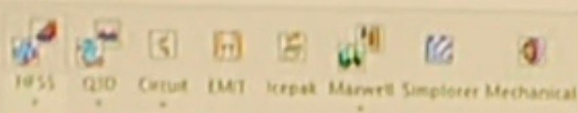
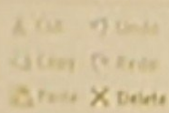


Altium Electronics Desktop 2021 R2 - Project1

File Edit View Project Tools Window Help



Save Archive
Restore Archive



Update Definitions
Remove Unused Definitions
Edit Definitions

Project Variables
Datasets
Event Callbacks

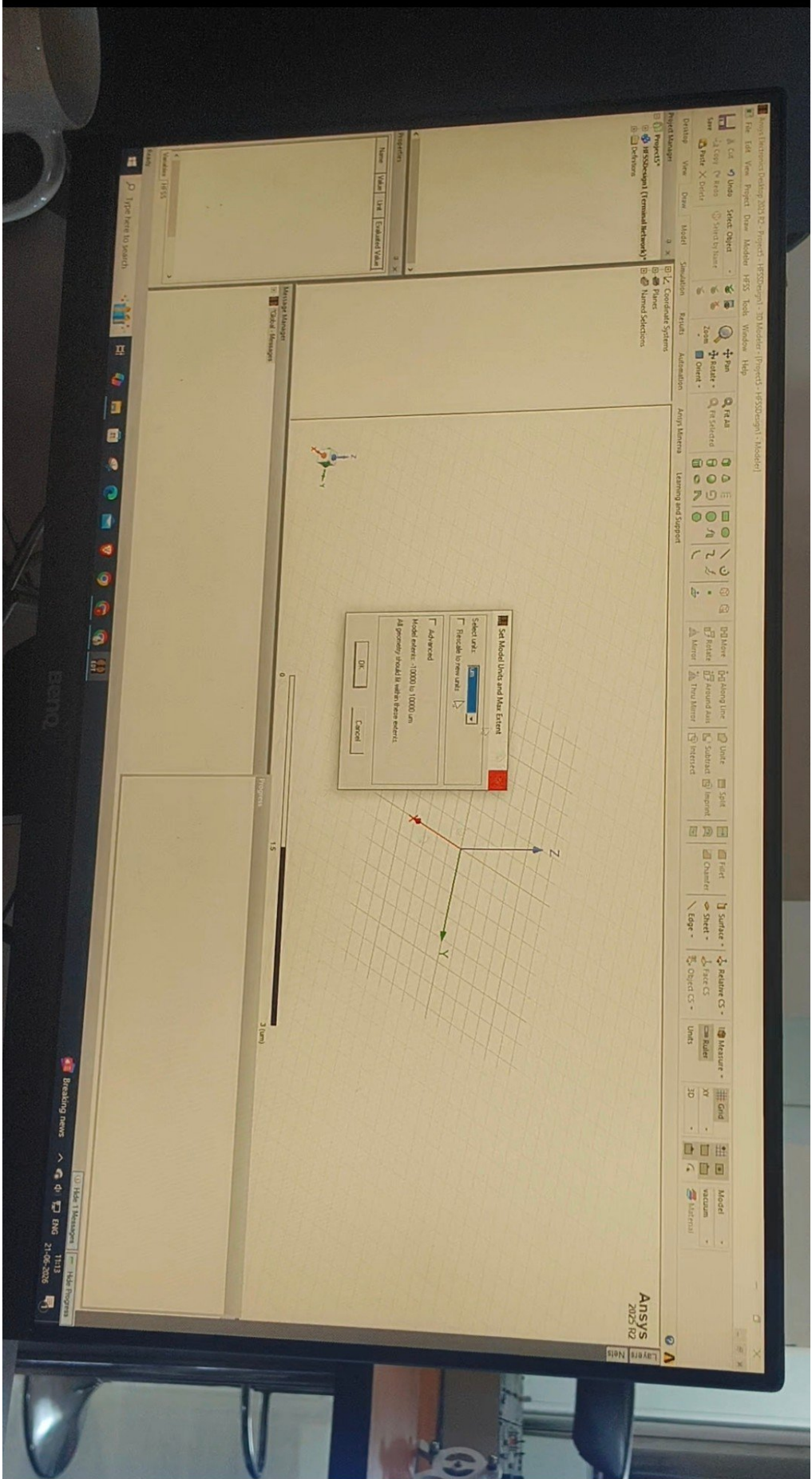
General Options

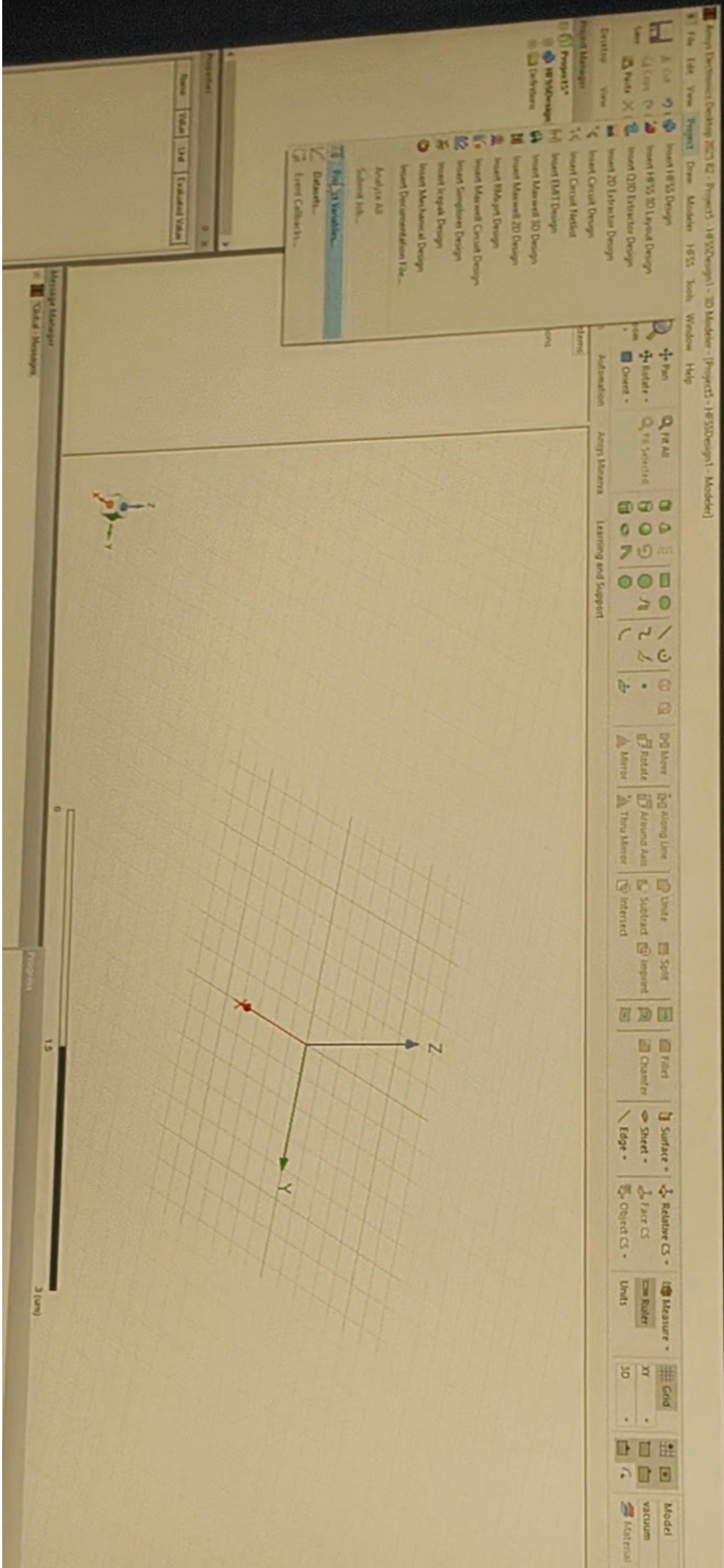
Desktop View Simulation Automation Ansys Minerva Learning and Support

Project Manager

Project1

1855
1855 3D Layout

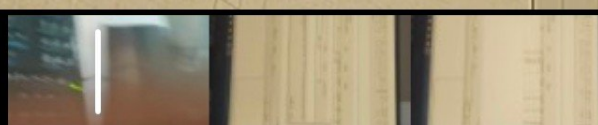
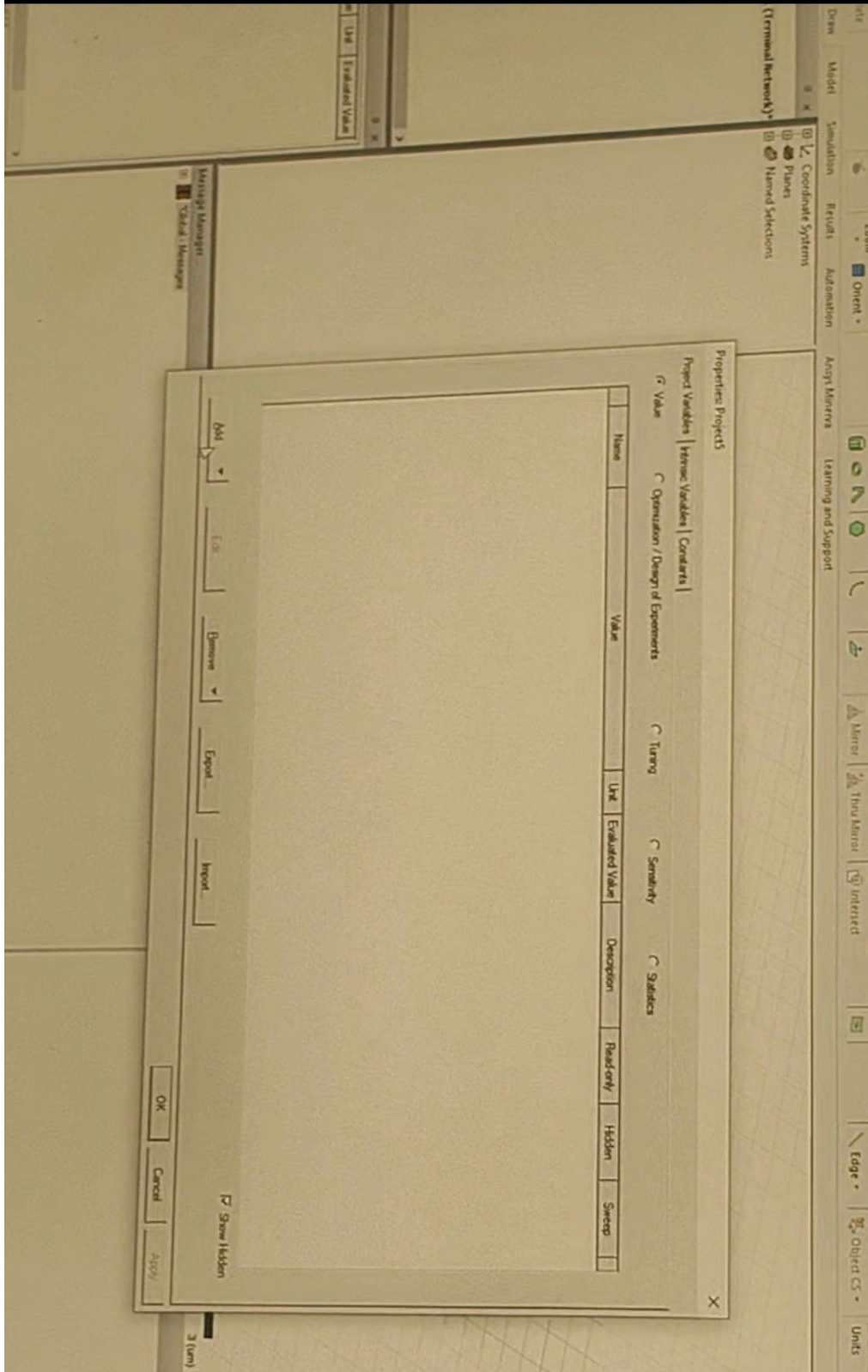






Today

11:36 am Uparwara



Share



Edit



Play



Delete

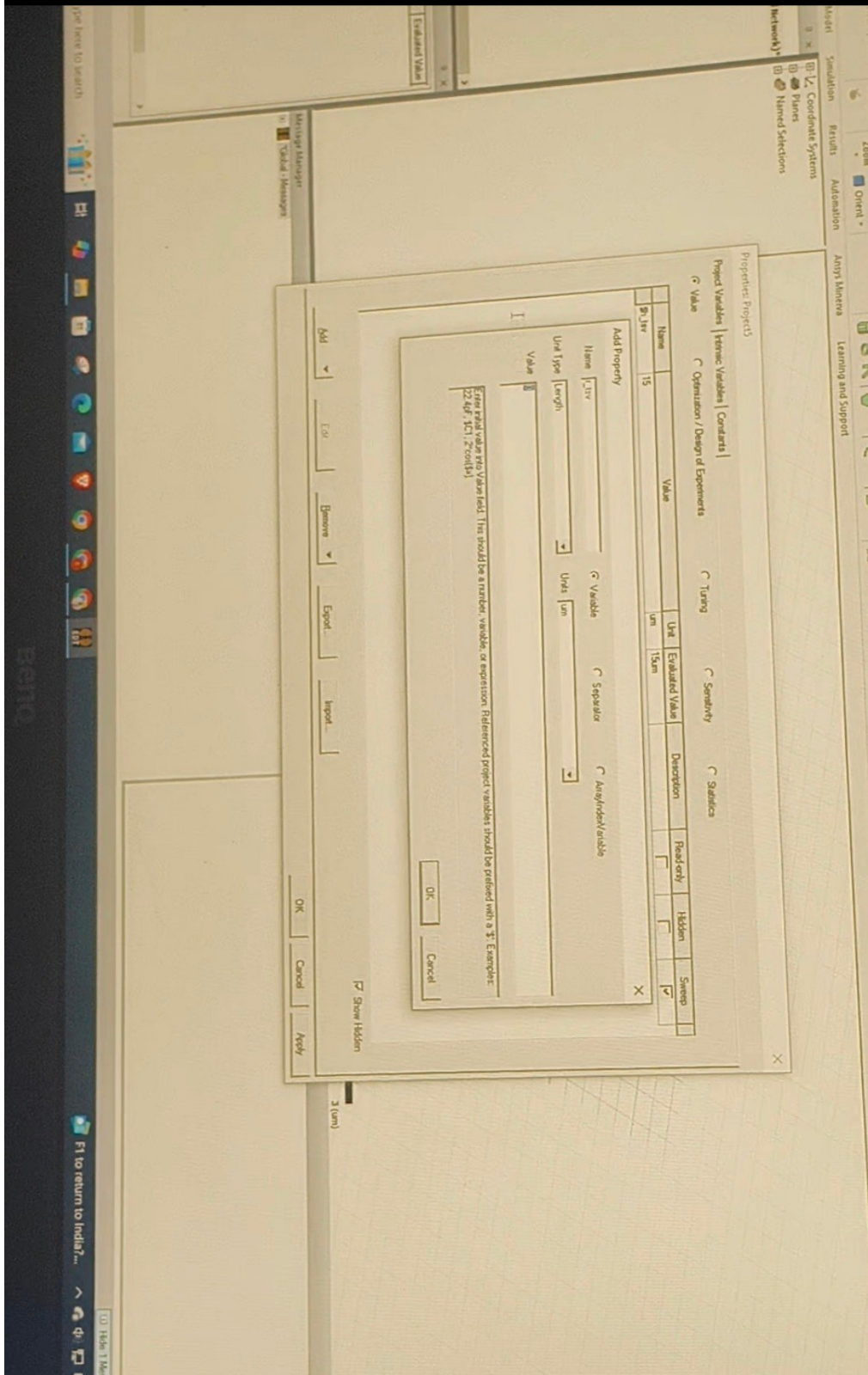


More



Today

11:36 am Uparwara



Share



Edit



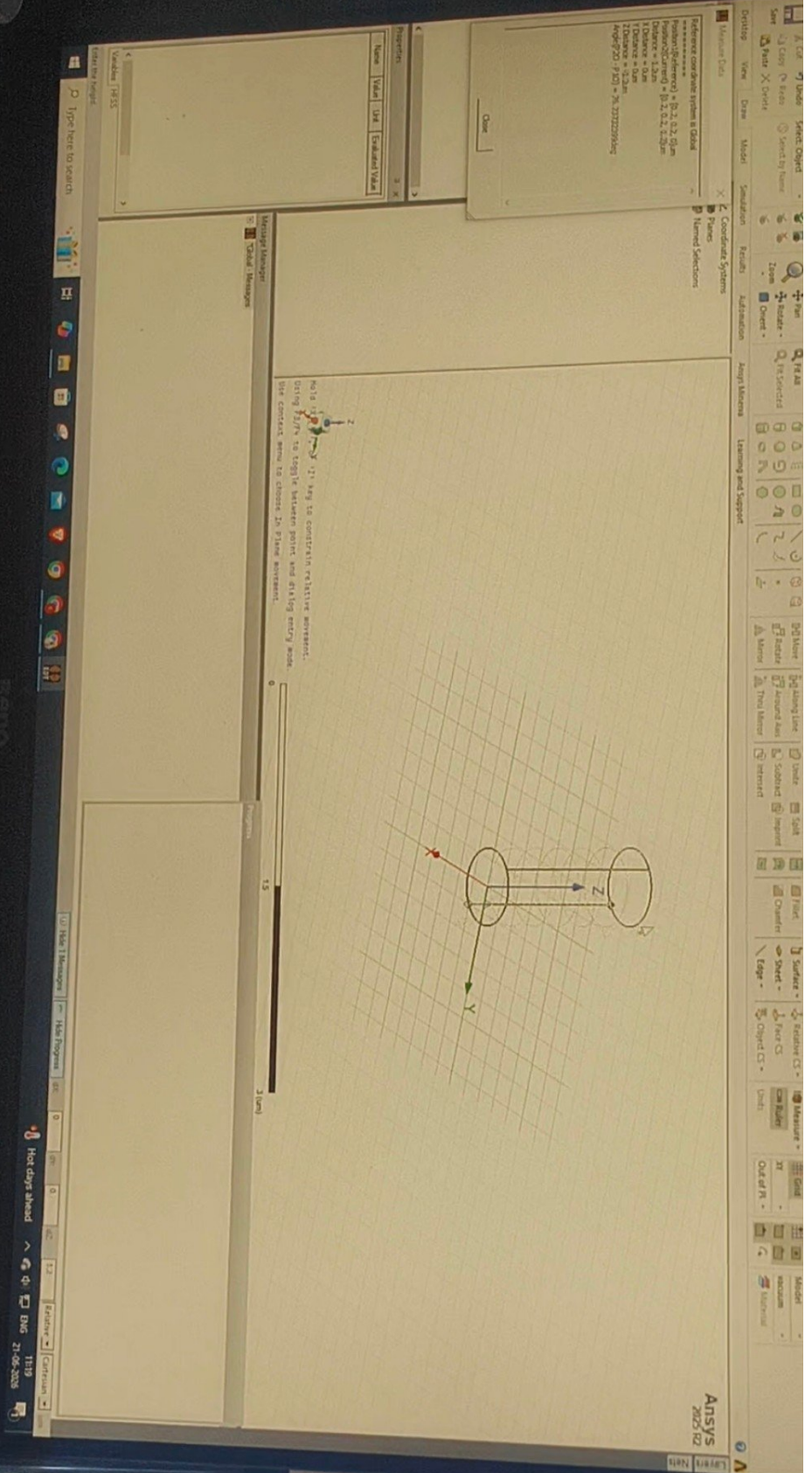
Play

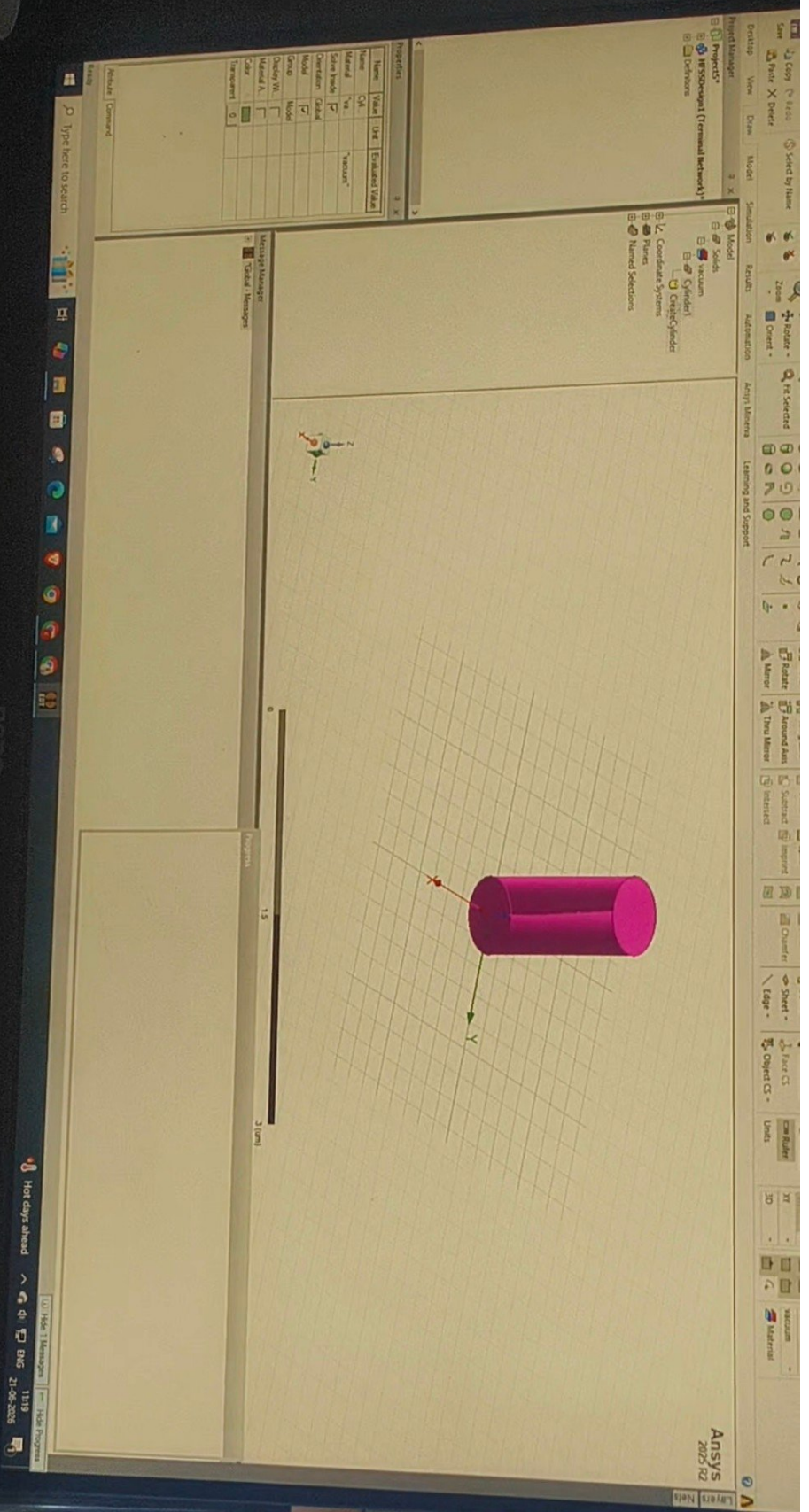


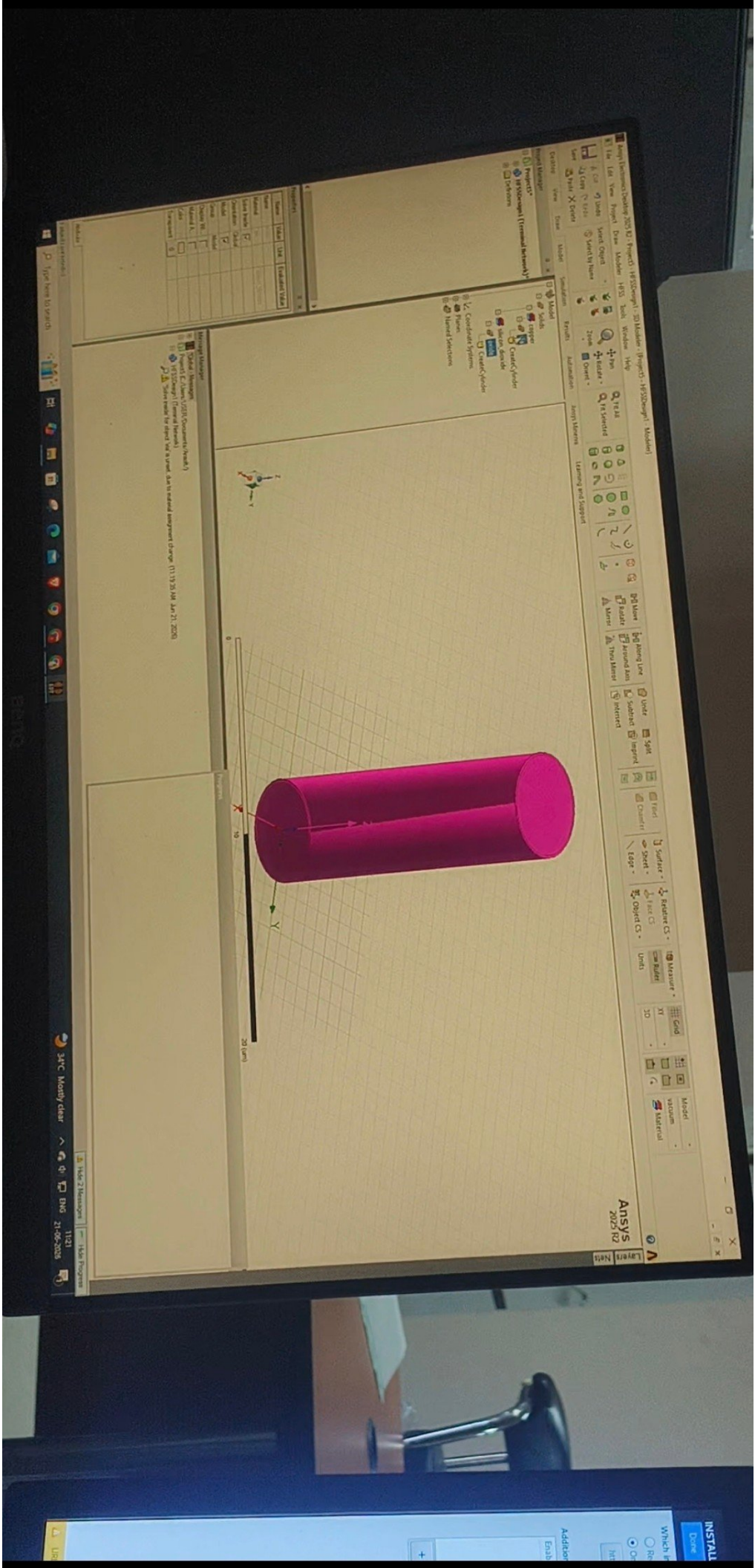
Delete



More







ANSYS 2023 R2

Tree Outline

- Project1
 - Geometry
 - Part1
 - Mesh
 - Mesh1
 - Solution
 - Solve
 - Results
 - Post1

Properties

Name	Type	Material	Element Type	Real Constant	Section	Coordinate System	Support	Load	Analysis Method
Part1	Part	Steel	SOLID208			CS1			Static

Modeling and Support

34°C Mostly clear 11:21 21-05-2023

INSTALL

Which if

Done

Add

Enab



Subtract

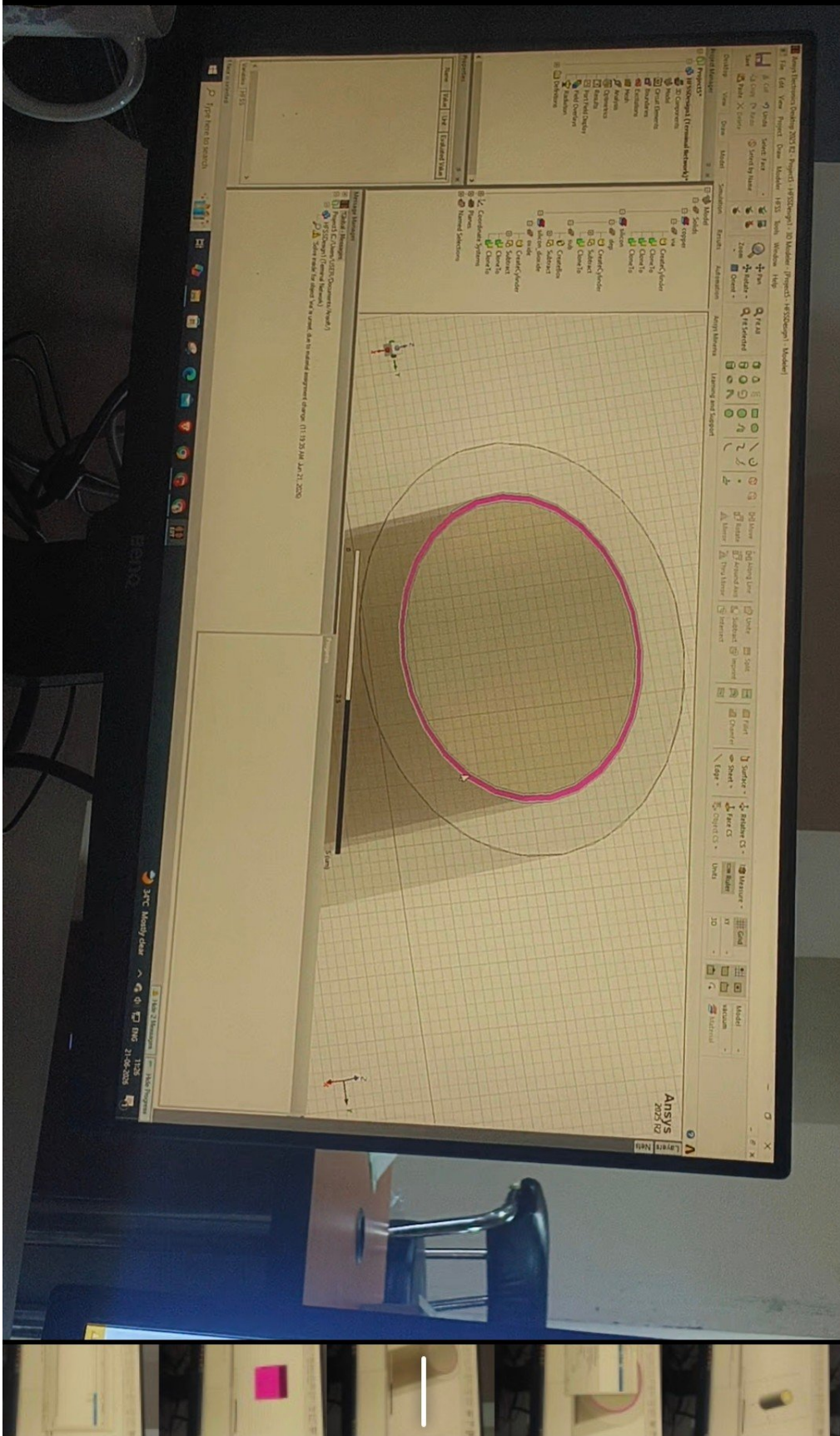
Blank Part	Tool Part
Order	Order

☒ Close tool objects before operation



Properties

< Today
11:36 am Uparwara



Share



Edit



Play

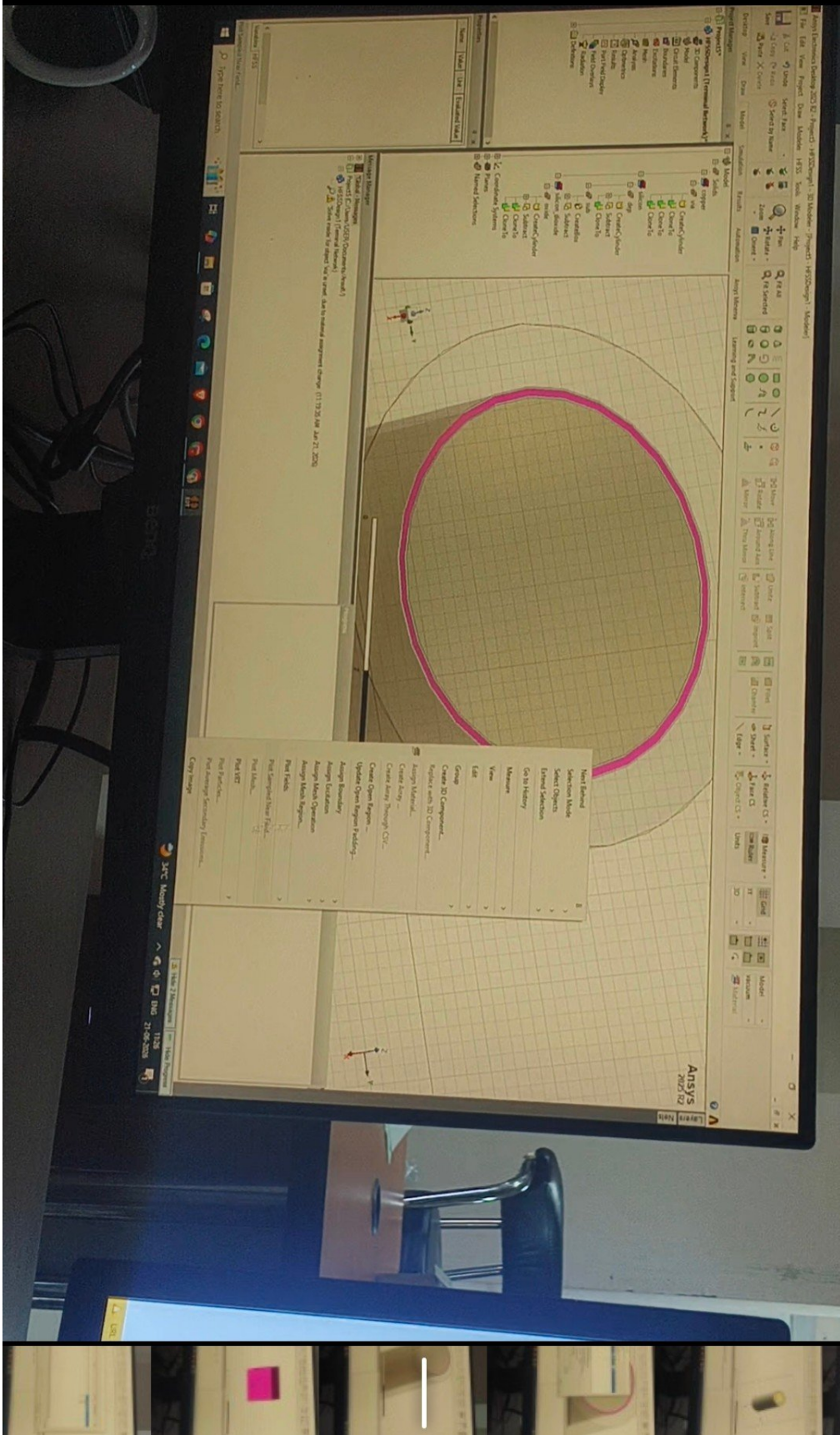


Delete



More

< Today
11:36 am Uparwara



Share



Edit



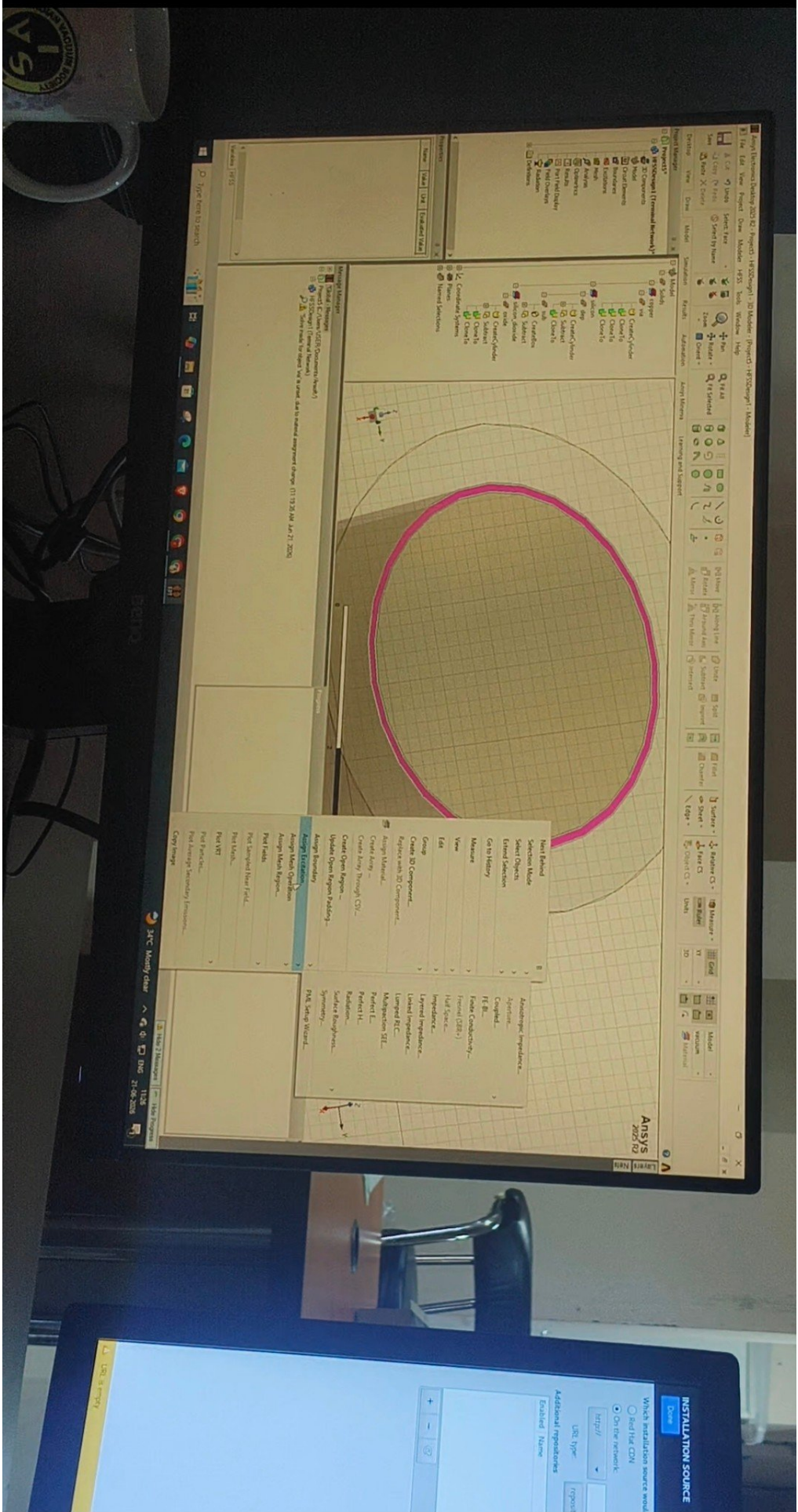
Play



Delete



More



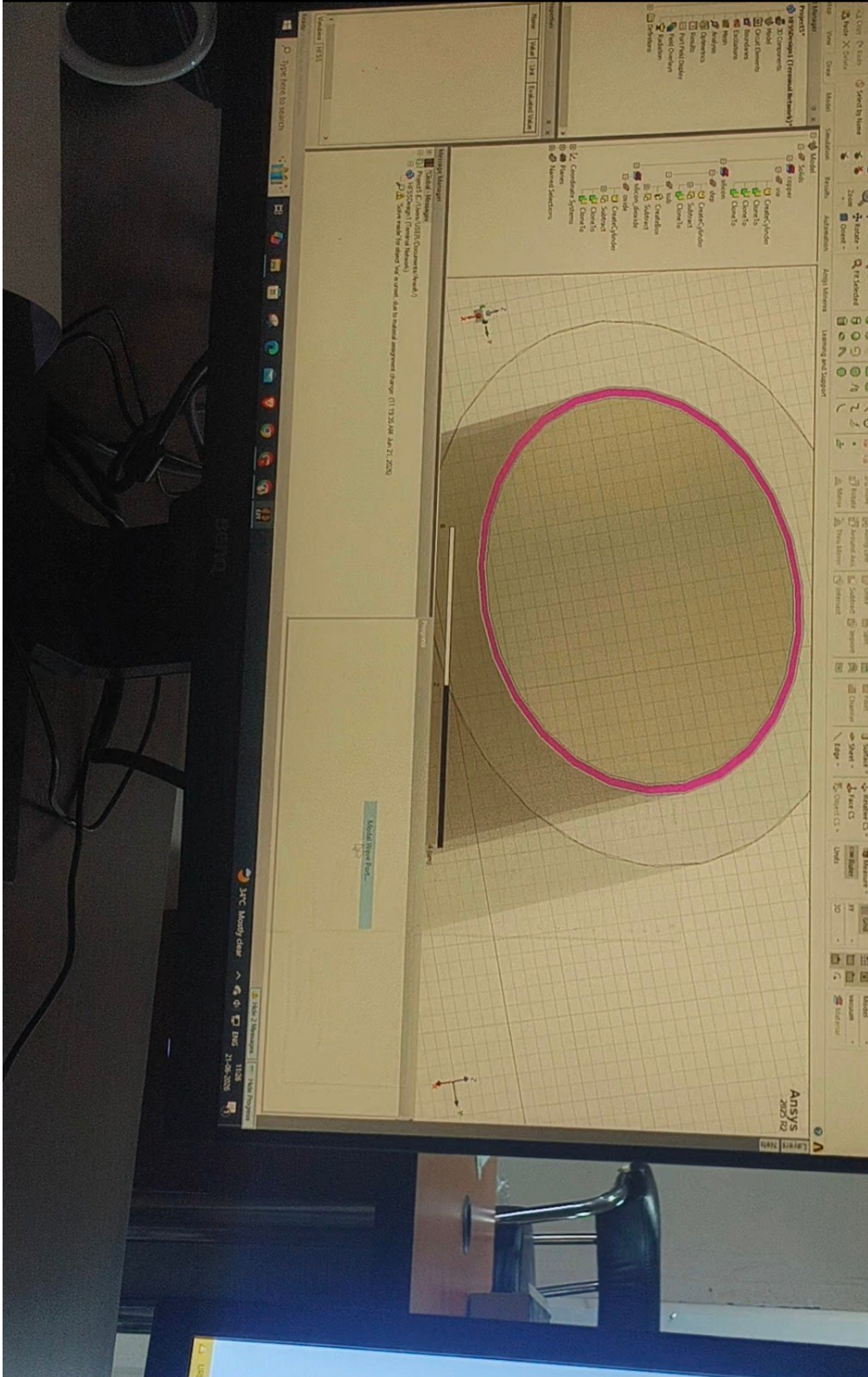
6:32

Vol 0.00 5G KB/s



Today

11:36 am Uparwara



Share



Edit



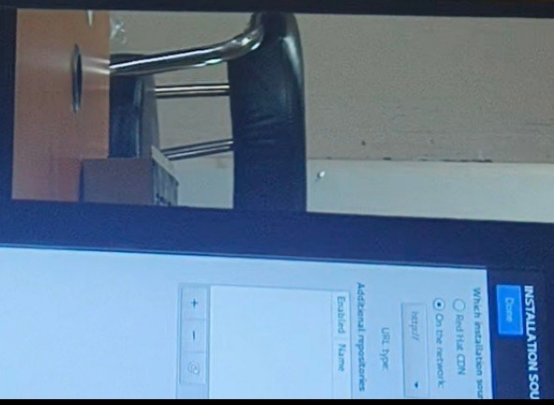
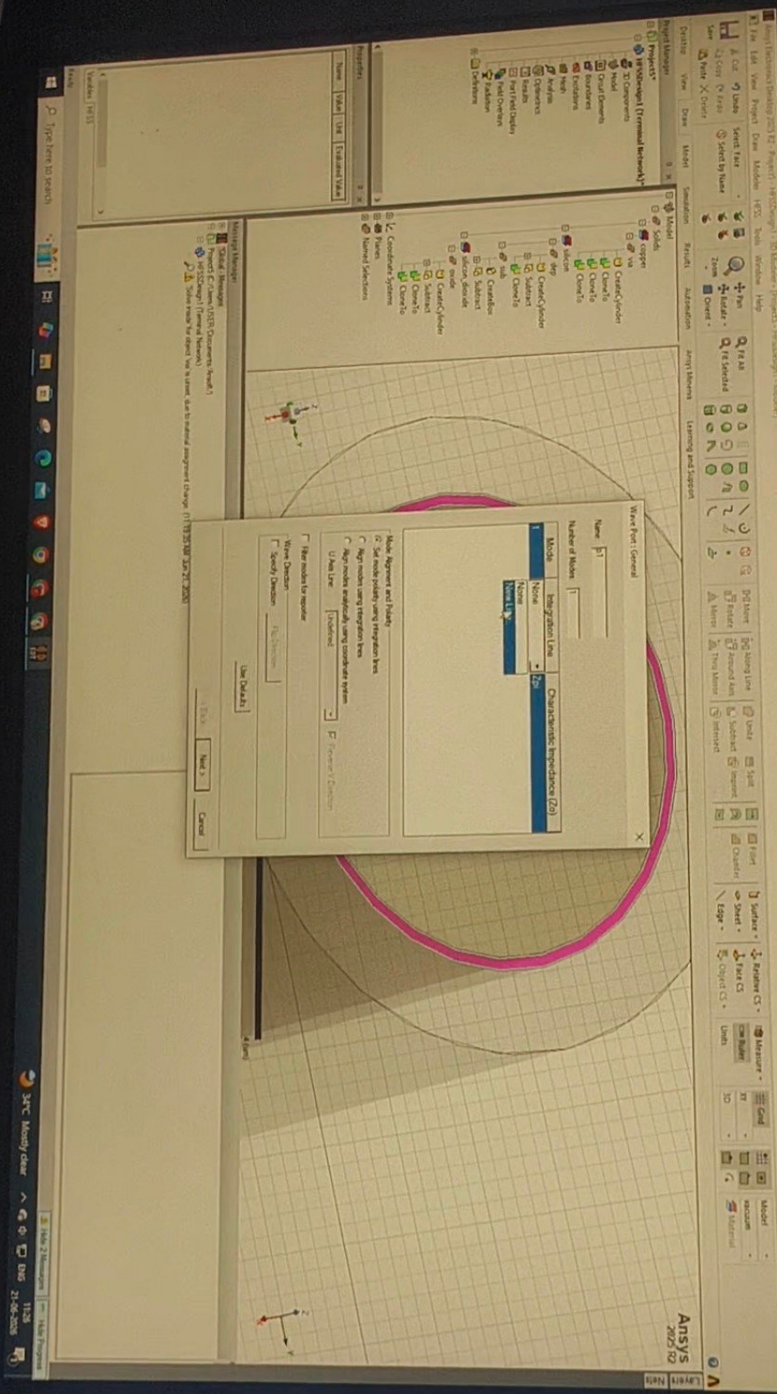
Play

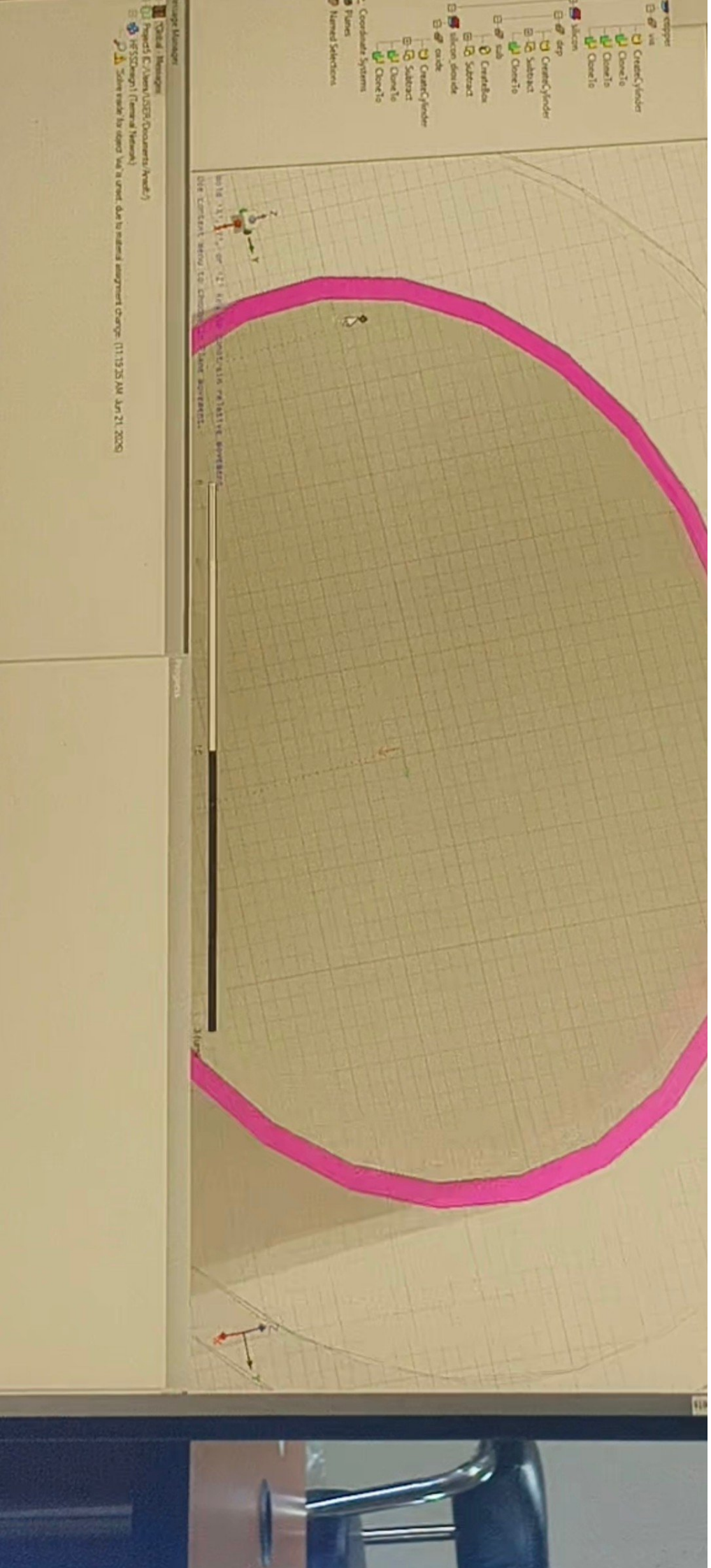


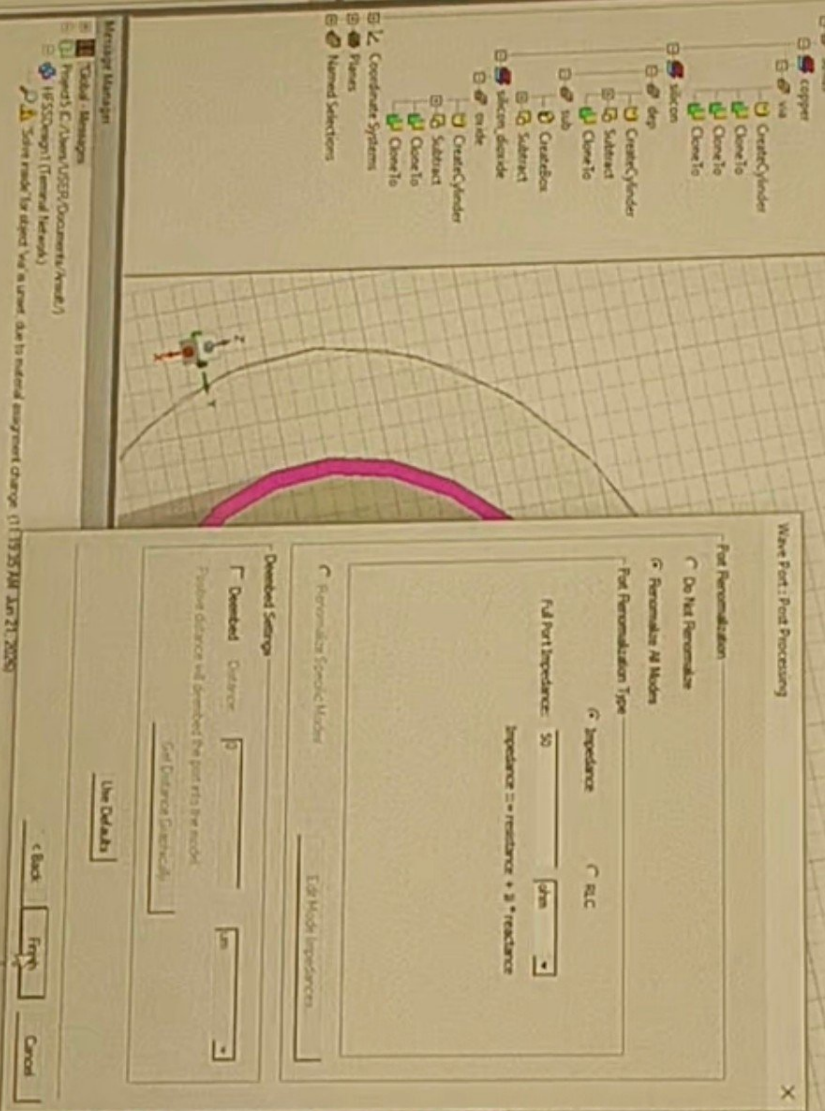
Delete



More





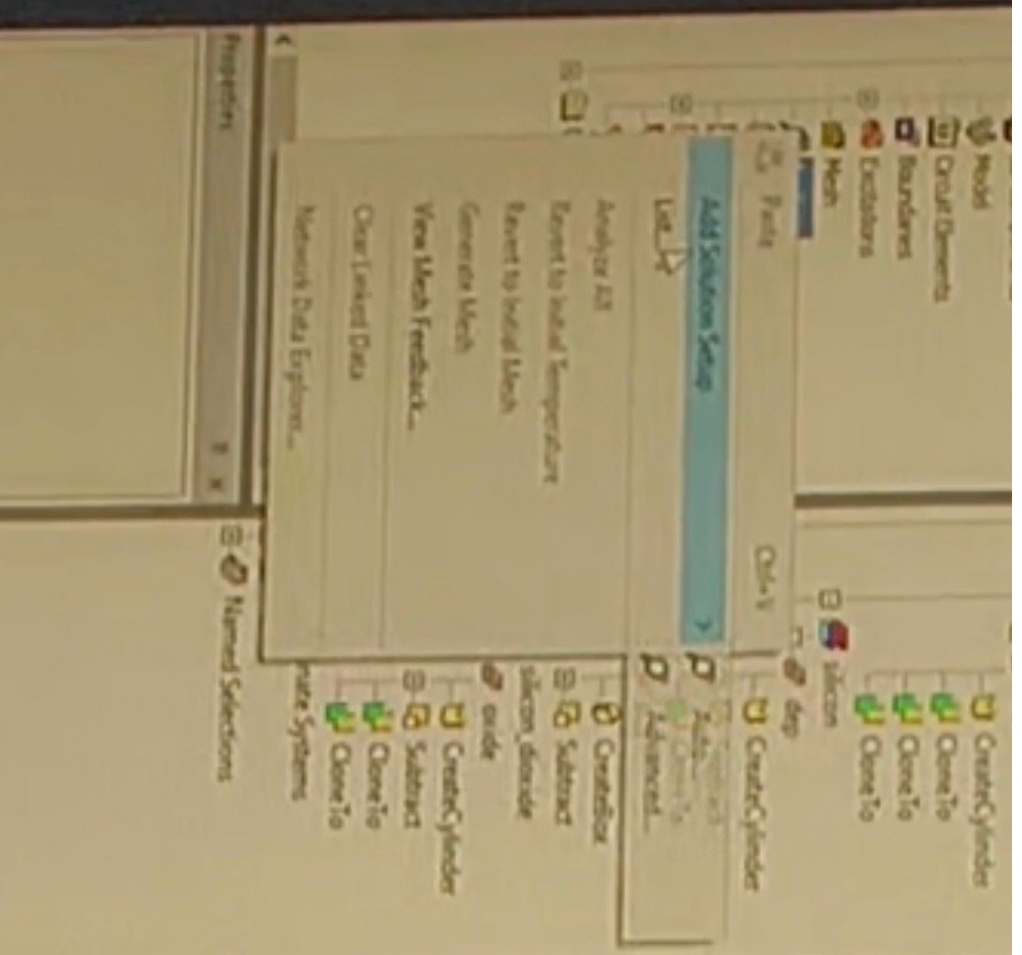


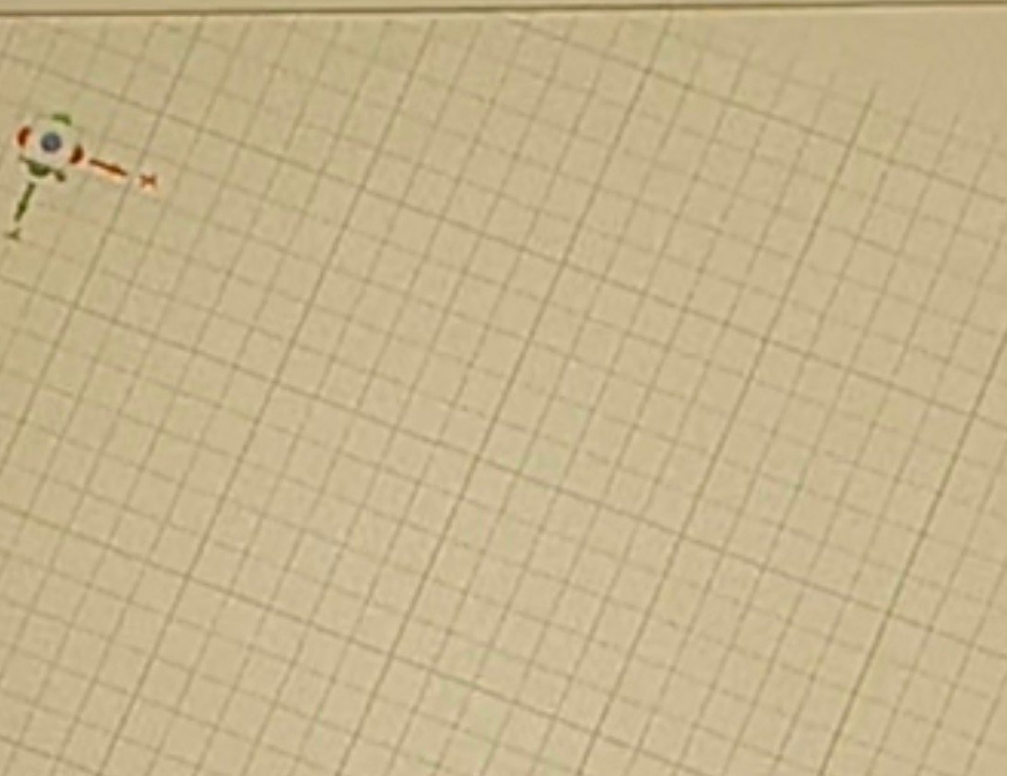
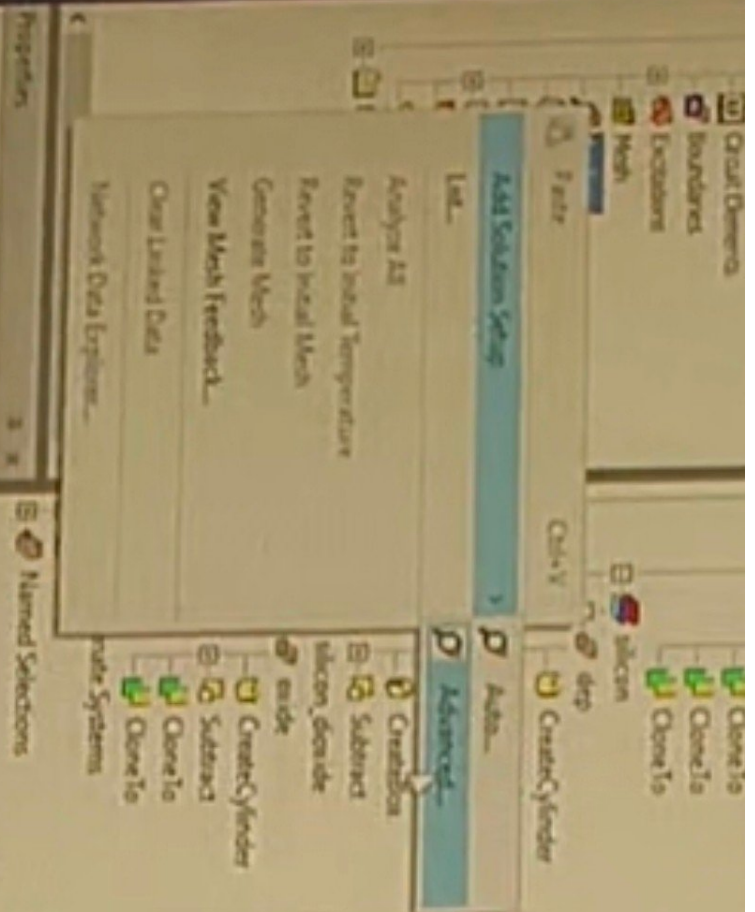
Messages Manager

Project3 E:\Users\USER\Documents\Project3

16:52:00 (Internal Network)

Save model for object 'Via' is saved. due to internal assignment change (11/19/2014 Jun 21, 2014)





- Scalls
 - copper
 - via
 - CreateCylinder
 - CloneTo
 - CloneTo
 - CloneTo
 - silicon
 - drag
 - CreateCylinder
 - Subtract
 - CloneTo
 - sub
 - CreateBox
 - Subtract
 - silicon_device
 - outside
 - CreateCylinder
 - Subtract
 - CloneTo
 - Coordinate Systems
 - Planes
 - Named Selections

Message Manager

- Global Messages
 - Project C:\Users\USER\Documents\Ansys\
 - HFS50comp1 (General Message)
 - Save Fields for object 'net' is unset, due to internal assignment change. (11:35 AM, Jan 21, 2025)
 - Permanently the port impedance will not repeat the field results if the 'Include Post Processing Effects' is checked in EIS Sources. (11:27:22 AM, Jan 21, 2025)
 - Permanently the port impedance will not repeat the field results if the 'Include Post Processing Effects' is checked in EIS Sources. (11:28:14 AM, Jan 21, 2025)

Edit Frequency Sweep

General | Interpolation | Details

Sweep Name: Sweep ☒ Enabled

Sweep Type: Interpolating ☐ Use QSD to solve the DC port (Beta)

Frequency Sweeps (4:1 ports defined)

Distribution	Start	End	Points	QSD
Linear Count	100Hz	1000Hz	1000	

- 3D Fields Save Options

☐ Save Fields (N Base Freq)

☐ Save Incident Fields only

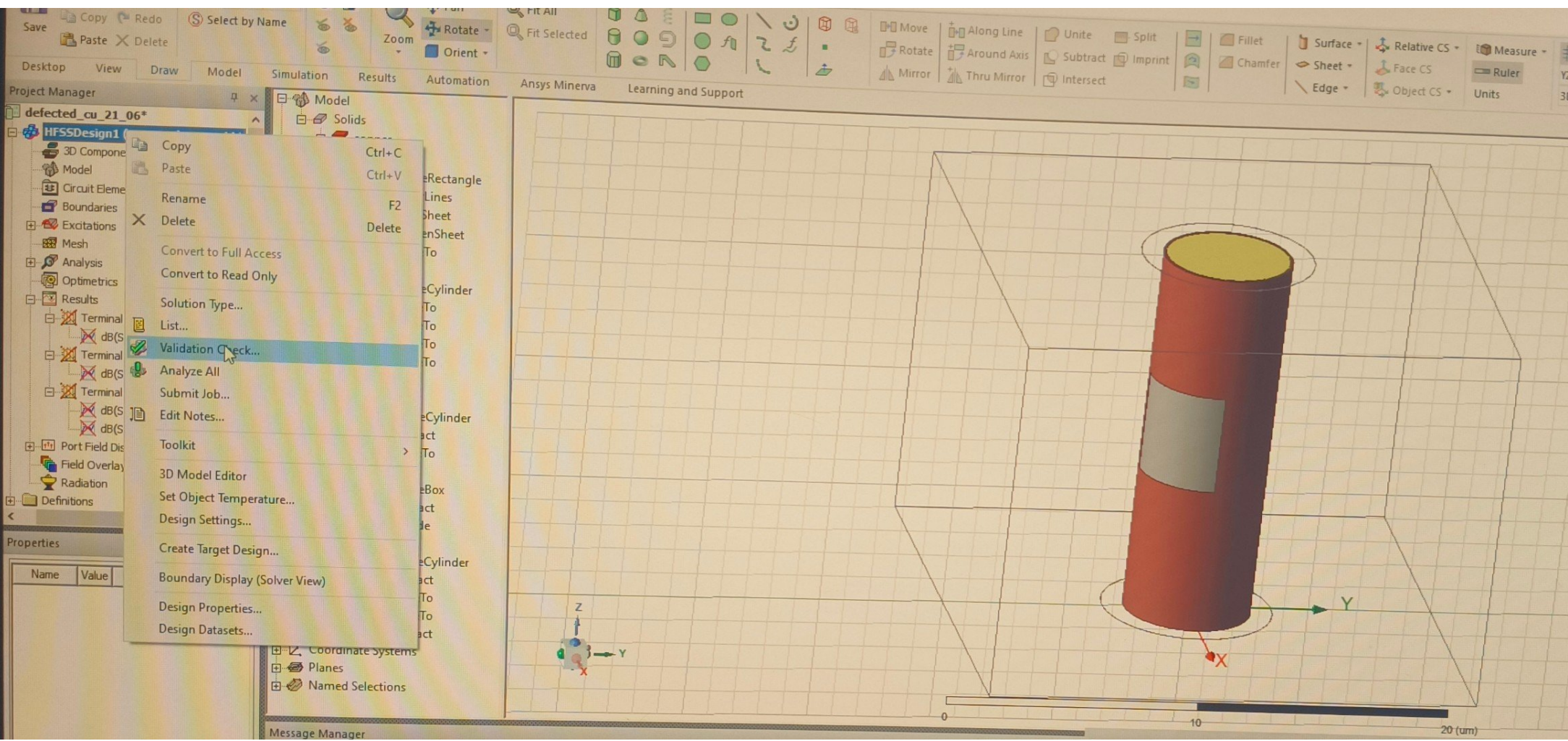
Time Domain Calculations

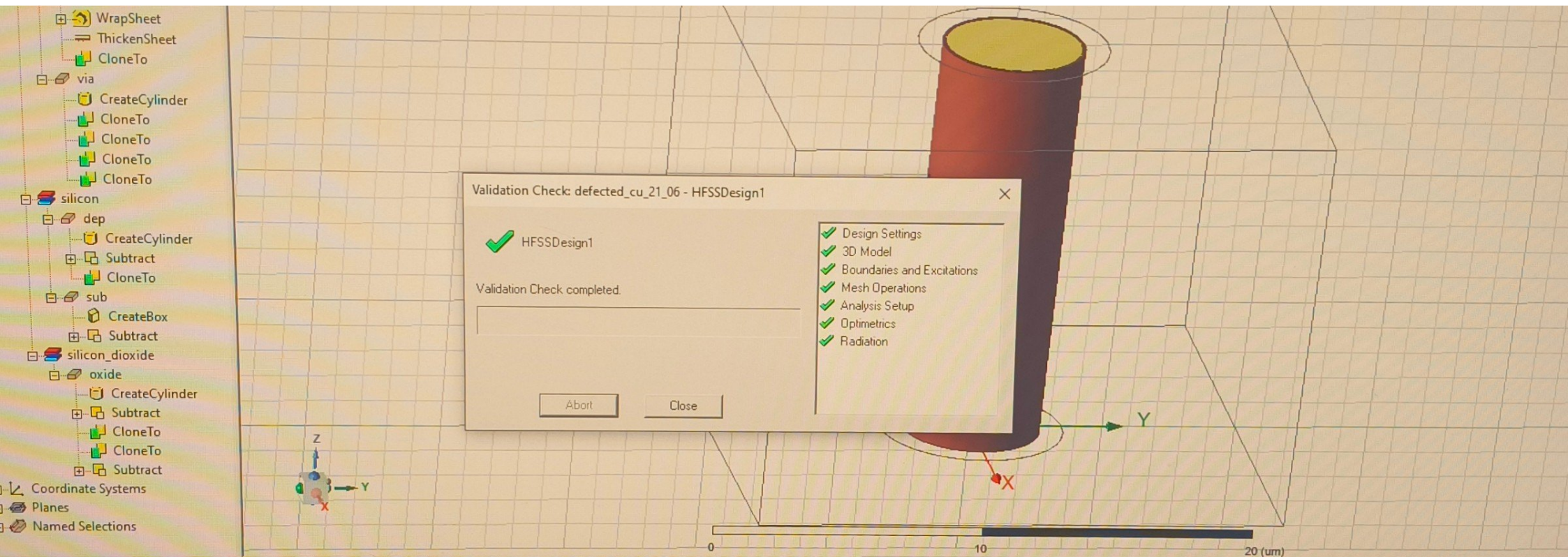
☒ S Matrix Only Solve

☐ Auto

☐ Manual - Allow for frequencies above

100Hz





Message Manager

- HFSSDesign1 (Terminal Network)
- 'Solve inside' for object 'via' is unset, due to material assignment change. (11:19:35 AM Jun 21, 2026)
- Renormalizing the port impedance will not impact the field results if the 'Include Port Post Processing Effects' is disabled in Edit Sources. (11:27:22 AM Jun 21, 2026)
- Renormalizing the port impedance will not impact the field results if the 'Include Port Post Processing Effects' is disabled in Edit Sources. (11:28:14 AM Jun 21, 2026)
- An interpolating frequency sweep with 100 points has been started using HFSS - Solving Distributed. (11:30:38 AM Jun 21, 2026)
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- Normal completion of simulation on server: Local Machine. (11:30:38 AM Jun 21, 2026)
- HFSSDesign1: Solutions have been invalidated. Undo to recover. (11:34:19 AM Jun 21, 2026)

Progress

Google Chrome



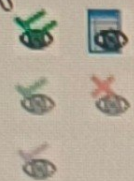
How to Design Power
an FPGA/SoC PCB LS

Electronics Desktop 2025 R2 - defected_cu_21_06 - HFSSDesign1 - 3D Modeler - [defected_cu_21_06 - HFSSDesign1 - Modeler]

File View Project Draw Modeler **HFSS** Tools Window Help

Cut Undo
Copy Redo
Paste Delete

Select: Face
Select by Name



Zoom

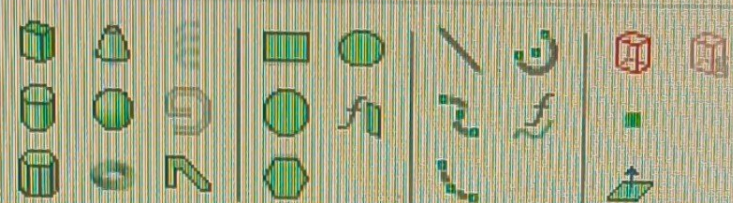
Pan

Rotate

Orient

Fit All

Fit Selected



View

Draw

Model

Simulation

Results

Automation

Ansys Minerva

Learning and Support

Project Manager



defected_cu_21_06*

Design1 (Terminal Network)*

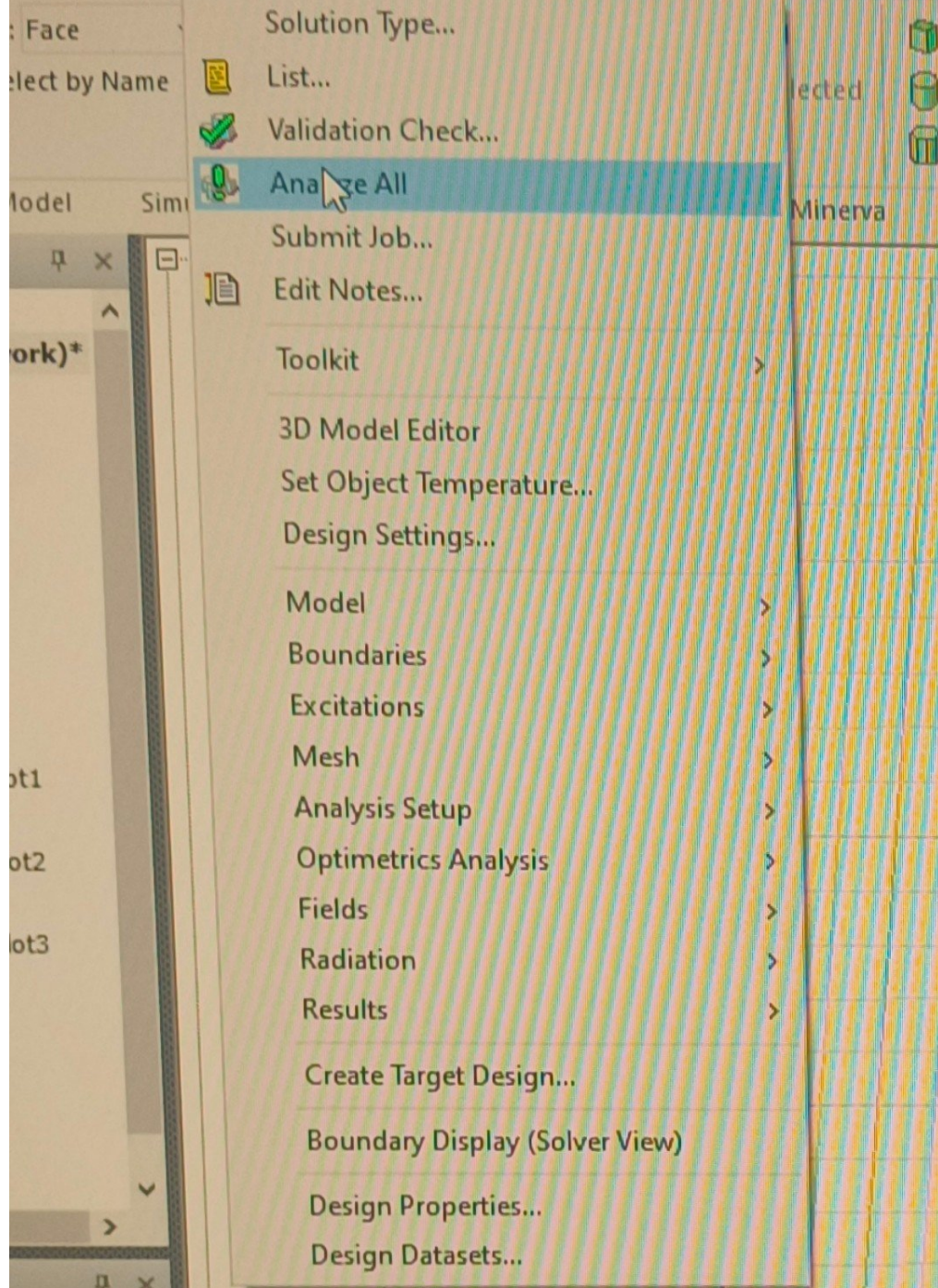
3D Components

Model

Model
Solids
copper
defect

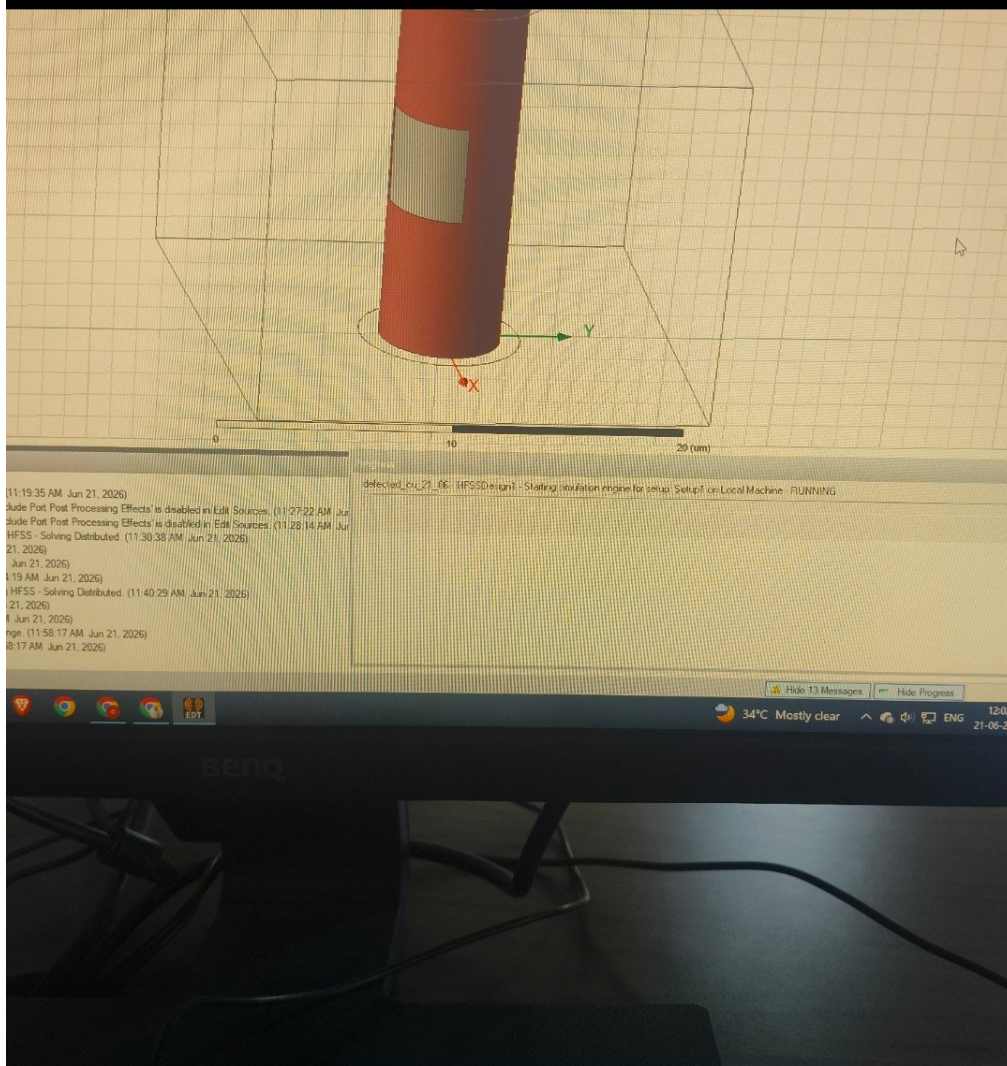
- defected_cu_21_06 - HFSSDesign1 - 3D Modeler - [defected_cu_21_06 - H

w Modeler HFSS Tools Window Help



ated Value

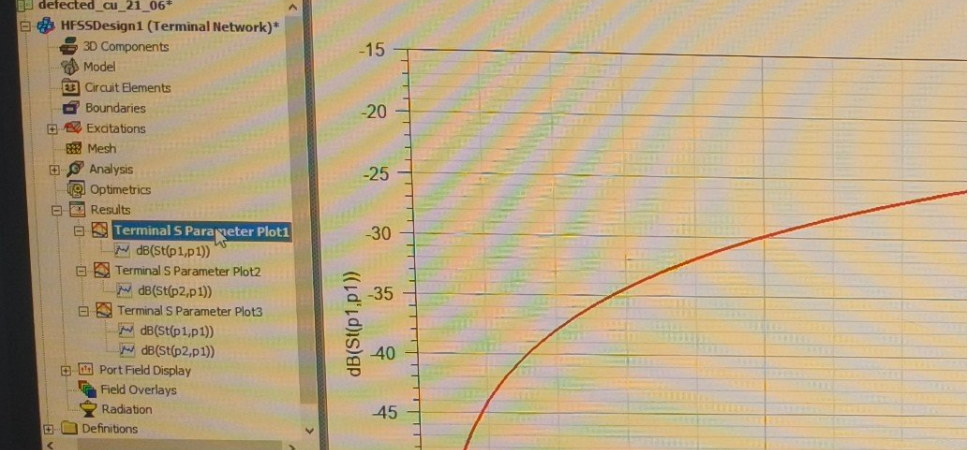
- CreateCylinder
- +..... Subtract
- CloneTo
- CloneTo
- +..... Subtract
- +..... Coordinate Systems



Ansys Electronics Desktop 2025 R2 - defected_cu_21_06 - HFSSDesign1 - Terminal S Parameter Plot1 - SOLVED - [defected_cu_21_06 - HFSSDesign1 - Termin

File Edit View Project Report2D HFSS Tools Window Help
Save Cut Undo Copy Paste X Delete Docking Windows Layouts Zoom Zoom Area Zoom Previous Pan Fit All Fit Selected Animate Copy Image

Desktop View Simulation Results Automation Ansys Minerva Learning and Support



Properties

Name	Value
Name	Terminal S Parameter Plot1
Report Type	Terminal Solution Data
Display Ty...	Rectangular Plot
dB(S(p1,p...	Freq. dB(S(p1,p1))

Message Manager

- HFSSDesign1 (Terminal Network)
 - Solve inside for object 'via' is unset, due to material assignment change. (11:19:35 AM Jun 21, 2026)
 - Renormalizing the port impedance will not impact the field results if the 'Include Port Post Processing E
 - Renormalizing the port impedance will not impact the field results if the 'Include Port Post Processing E
 - An interpolating frequency sweep with 100 points has been started using HFSS - Solving Distributed. (
 - Interpolating frequency sweep complete. Converged. (11:30:38 AM Jun 21, 2026)
 - Normal completion of simulation on server: Local Machine. (11:30:38 AM Jun 21, 2026)
 - HFSSDesign1: Solutions have been invalidated. Undo to recover. (11:34:19 AM Jun 21, 2026)
 - An interpolating frequency sweep with 100 points has been started using HFSS - Solving Distributed. (
 - Interpolating frequency sweep complete. Converged. (11:40:29 AM Jun 21, 2026)
 - Normal completion of simulation on server: Local Machine. (11:40:29 AM Jun 21, 2026)
 - Solve inside for object 'defect' is unset, due to material assignment change. (11:58:17 AM Jun 21, 20
 - HFSSDesign1: Solutions have been invalidated. Undo to recover. (11:58:17 AM Jun 21, 2026)
 - An interpolating frequency sweep with 100 points has been started using HFSS - Solving Distributed. (

Type here to search

6:50

Vol 0.00 5G
LTE KB/s

HFSSDesign1 - 3D Modeler - [defectcylinder_21_06 - HFSSDesign1 - Modeler]

File Tools Window Help



Model Results Automation Ansys Minerva Learning and Support

Model

Solids

copper

via

CreateCylinder

CloneTo

CloneTo

CloneTo

CloneTo

silicon

dep

CreateCylinder

Subtract

CloneTo

sub

CreateEllBox

Subtract

silicon_dioxide

inside

CreateCylinder

Subtract

CloneTo

CloneTo

Sheets

Unassigned

Rectangle1

CreateRectangle

CoverLines

WrapSheet

Coordinate Systems

Planes

Named Selections

Properties: defectcylinder_21_06 - HFSSDesign1 - Modeler

Command

Name	Value	Unit	Evaluated Value
Command	CreateRectangle		
Coordinate Sys	Global		
Position	0,20,10		
Axis	X	um	0um, 20um, 10um
YSize	3.87	um	3.87um
ZSize	4	um	4um

Message Manager

Tcl/Tk - Messages

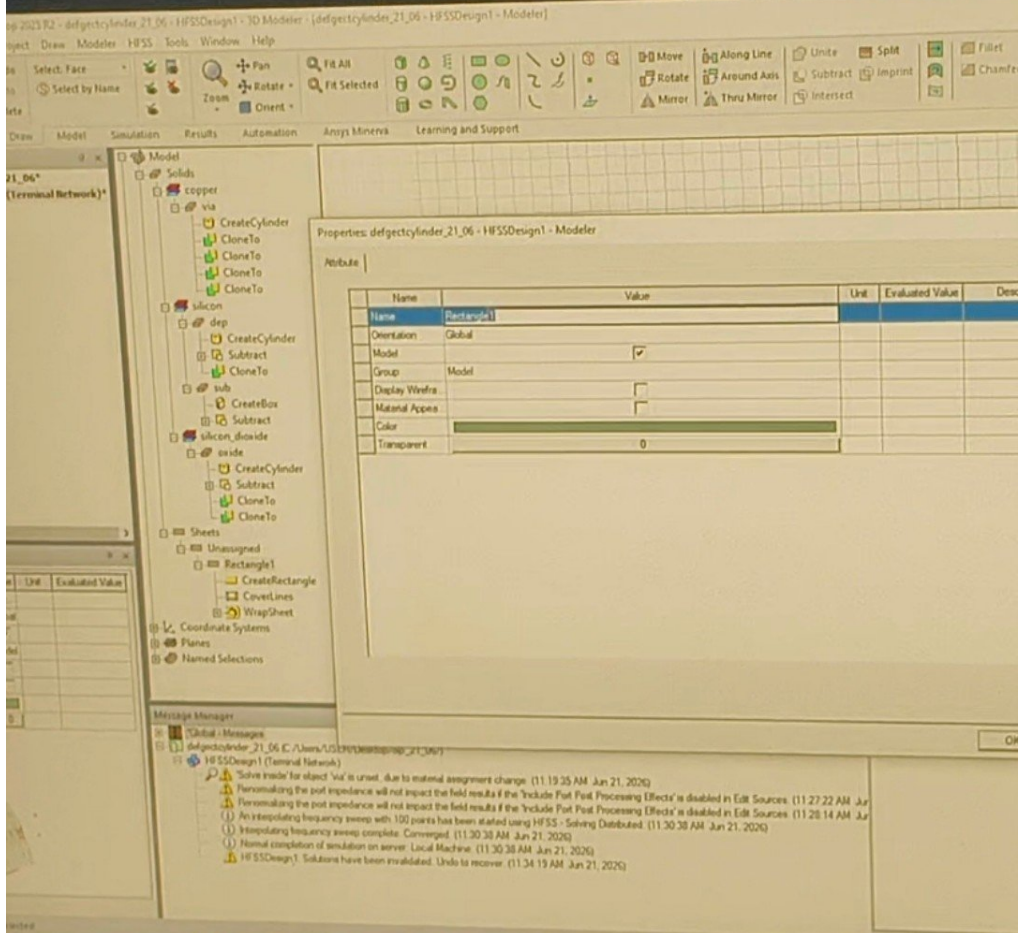
defectcylinder_21_06 (C:\Users\USP\Desktop\defectcylinder_21_06)

HFSSDesign1 (Terminal Network)

- Solve mode for object 'via' is unset, due to material assignment change. (11:19:35 AM Jun 21, 2020)
- Personalizing the port impedance will not impact the field results if the 'Include Post Processing Effects' is disabled in Edit Sources. (11:27:22 AM Jun 21, 2020)
- An interpolating frequency sweep with 100 points has been started using HFSS - Solving Distributed. (11:30:14 AM Jun 21, 2020)
- HFSSDesign1: Solutions have been invalidated. Undo to recover. (11:34:19 AM Jun 21, 2020)

OK

Cancel

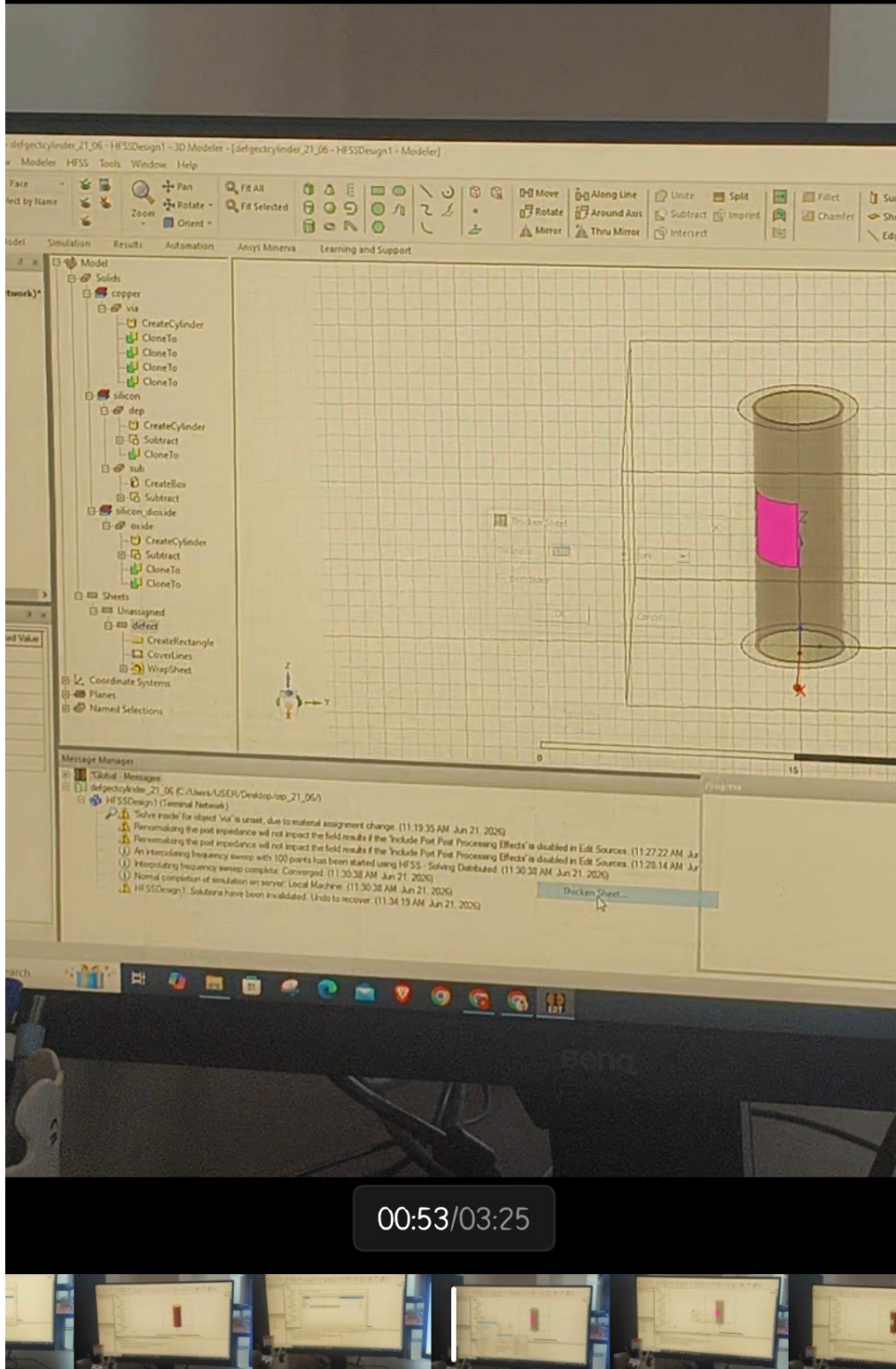


6:51



Today

11:42 am Uparwara



Share



Edit



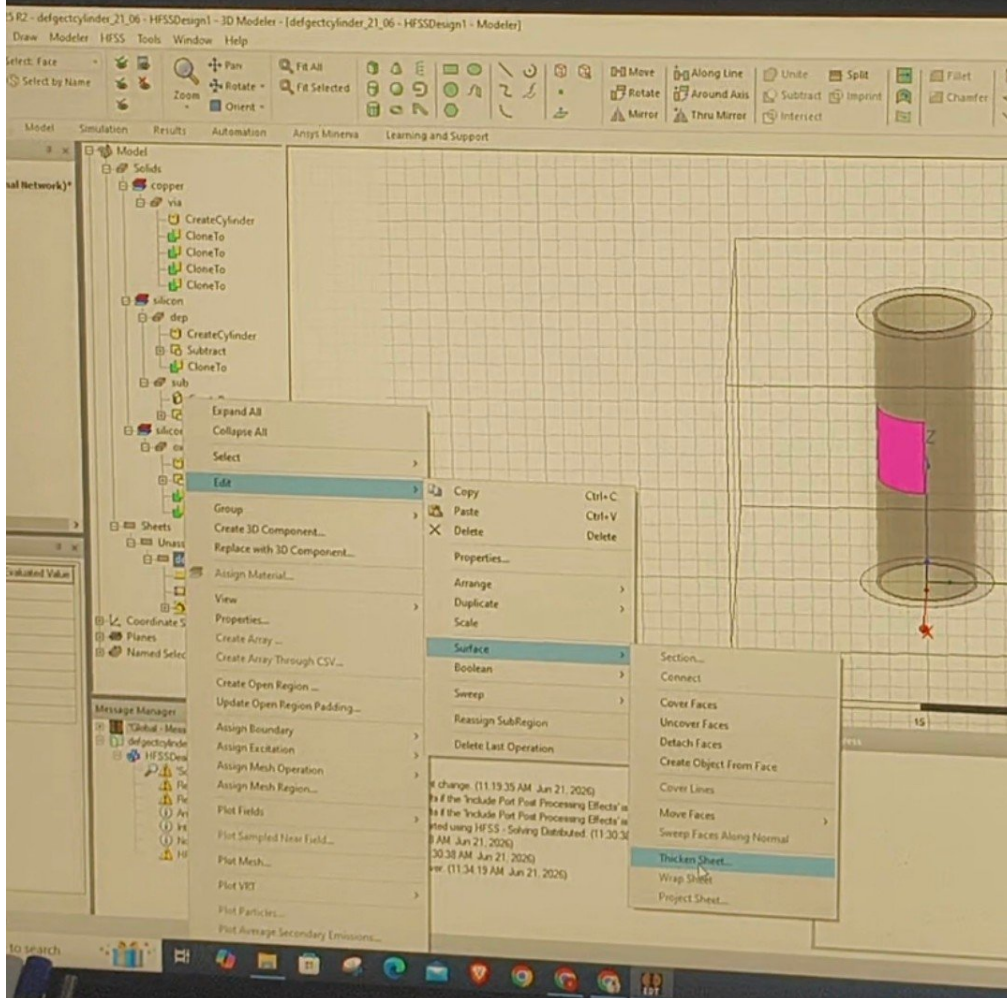
Pause



Delete



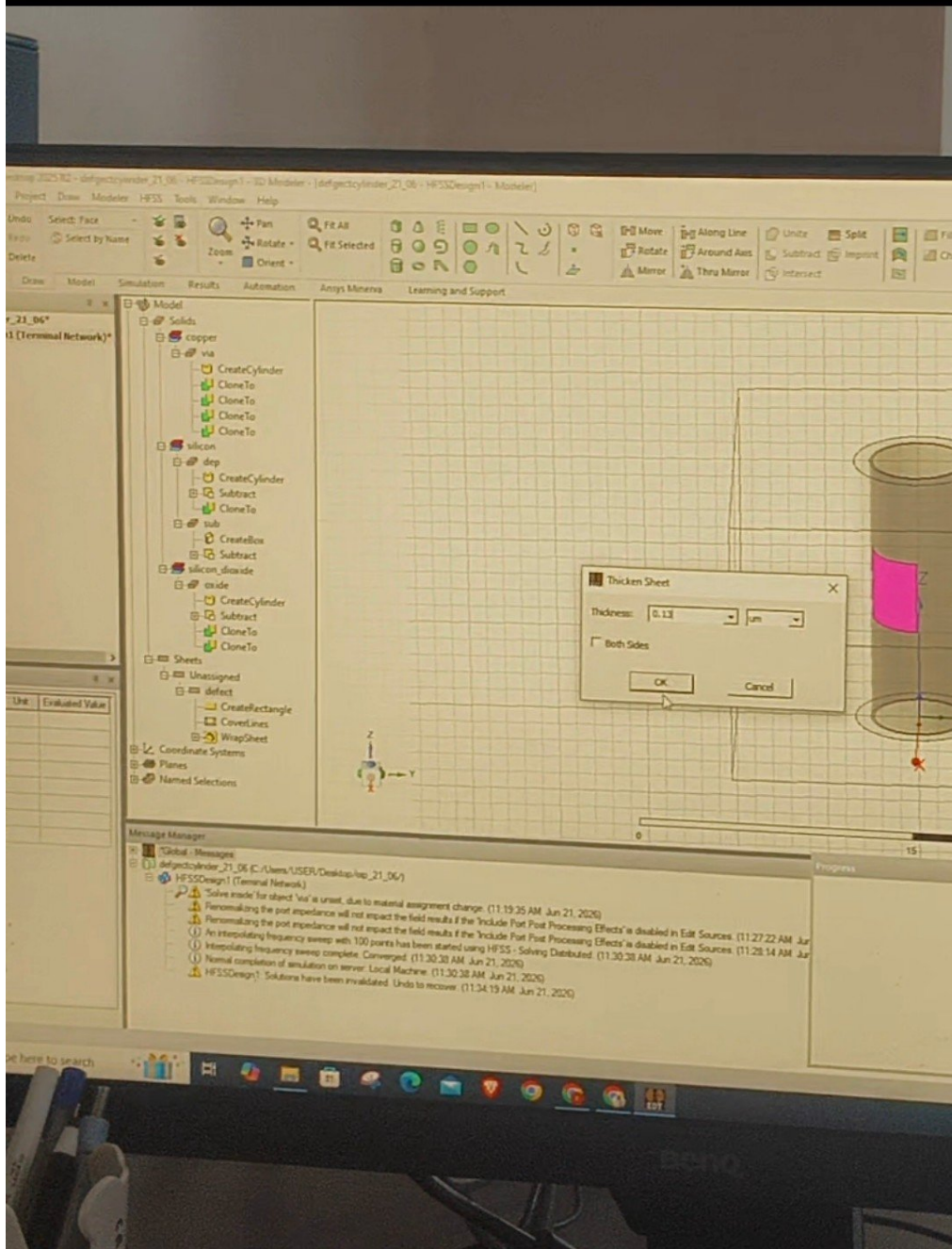
More



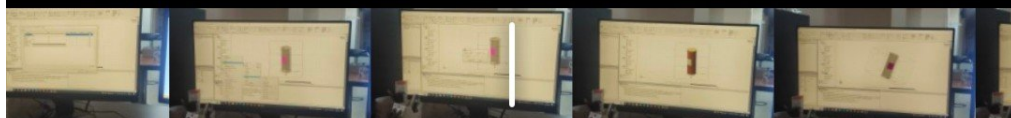
6:51

VoLTE 0.10 KB/s 5G

< Today
11:42 am Uparwara



01:13/03:25



Share



Edit



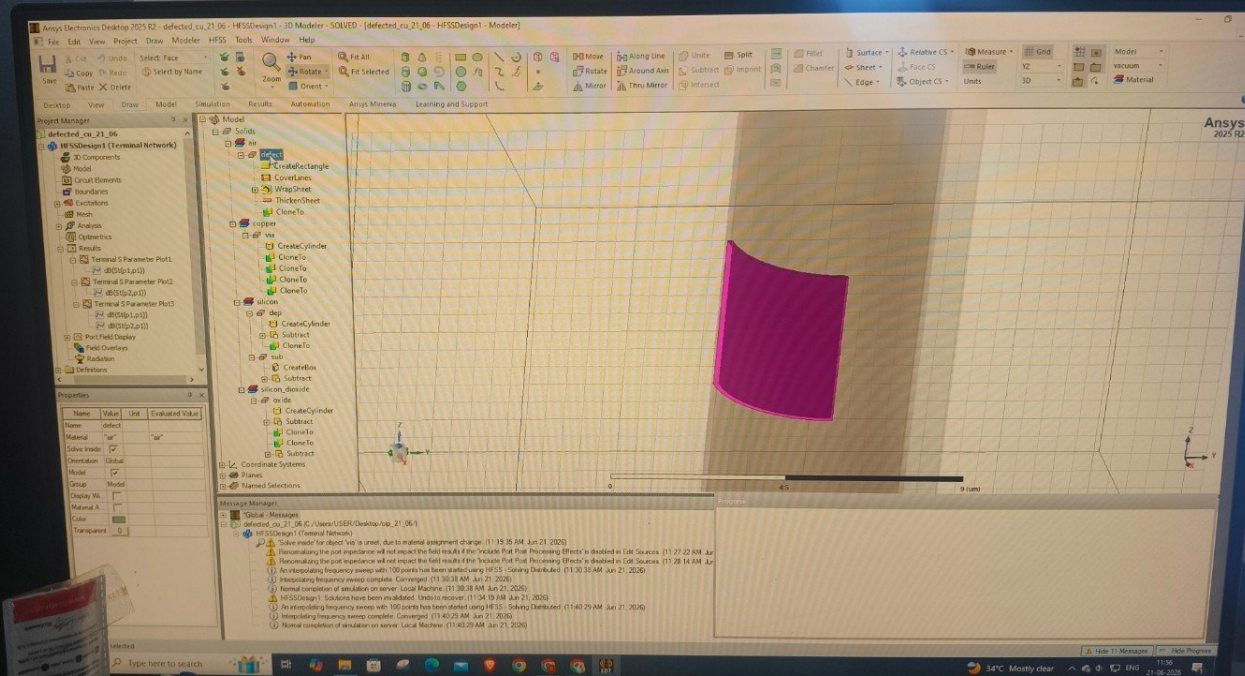
Pause

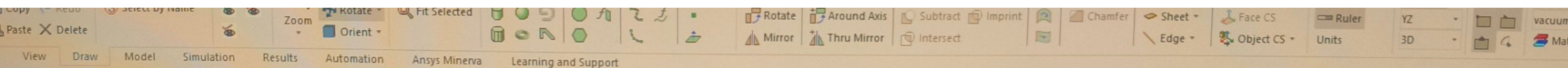


Delete



More





Modeler

Model

- defected_cu_21_06
- HFSSDesign1 (Terminal Network)
- 3D Components
- Model
- Circuit Elements
- Boundaries
- Excitations
- Mesh
- Analysis
- Optimetrics
- Results
- Terminal S Parameter Plot1
 - dB(St(p1,p1))
- Terminal S Parameter Plot2
 - dB(St(p2,p1))
- Terminal S Parameter Plot3
 - dB(St(p1,p1))
 - dB(St(p2,p1))
- Port Field Display
- Field Overlays
- Radiation
- Definitions

Model

- Solids
 - air
 - defect
 - CreateRectangle
 - CoverLines
 - WrapSheet
 - ThickenSheet
 - CloneTo
 - copper
 - via
 - CreateCylinder
 - CloneTo
 - CloneTo
 - CloneTo
 - silicon
 - dep
 - CreateCylinder
 - Subtract
 - CloneTo
 - sub
 - CreateBox
 - Subtract
 - silicon_dioxide
 - oxide
 - CreateCylinder
 - Subtract
 - CloneTo
 - CloneTo
 - Subtract
- Coordinate Systems
- Planes
- Named Selections

Message Manager

- *Global - Messages
- defected_cu_21_06 (C:/Users/USER/Desktop/oi_p_21_06/)
- HFSSDesign1 (Terminal Network)

Name	Value	Unit	Evaluated Value
defect			
Material	"air"		"air"
Solve Inside	☒		
Orientation	"air"		
Model	"copper"		
Group	"silicon_dioxide"		
Display Wireframe	☐		
Material Appearance			
Color			
Transparent	0		

Properties: defected_cu_21_06 - HFSSDesign1 - Modeler

Attribute

Name	Value	Unit	Evaluated Value	Description	Read-only
Name	defect				☐
Material	"air"		"air"		☐
Solve Inside	☒				☐
Orientation	"air"				☐
Model	"copper"				☐
Group	"silicon_dioxide"				☐
Display Wireframe	☐				☐
Material Appearance					☐
Color					☐
Transparent	0				☐

Show Hidden

OK Cancel Apply

defect

- CreateRectangle
- CoverLines
- WrapSheet
- ThickenSheet
- CloneTo

copper

- via
 - CreateCylinder
 - CloneTo
 - CloneTo
 - CloneTo
- silicon
 - dep
 - CreateCylinder
 - Subtract
 - CloneTo
 - sub
 - CreateBox
 - Subtract
- silicon_dioxide
 - oxide
 - CreateCylinder
 - Subtract
 - CloneTo
 - Subtract

Coordinate Systems

Planes

Named Selections

Unit	Evaluated Value
"air"	

Properties: defect_cu_21_06 - HFSSDesign1 - Modeler

Attribute

Name	Value	Unit	Evaluated Value	Description	Read-only
Name	defect				
Material	"air"		"air"		☐
Solve Inside	Edit...				☐
Orientation	"air"				☐
Model	"copper"				☐
Group	"silicon"				☐
Display Wirefra...	"silicon_dioxide"				☐
Material Appea...	"vacuum"				☐
Color					☐
Transparent	0				☐

Show Hidden

OK Cancel Apply

Message Manager

- Global - Messages
- defected_cu_21_06 (C:/Users/USER/Desktop/oi_p_21_06/)
 - HFSSDesign1 (Terminal Network)
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9 (um)

HFSSDesign1 (Terminal Network)

3D Components

- Model
- Circuit Elements
- Boundaries
- Excitations
- Mesh
- Analysis
- Optimetrics
- Results
- Terminal S Parameter Plot1
 - dB(St(p1,p1))
- Terminal S Parameter Plot2
 - dB(St(p2,p1))
- Terminal S Parameter Plot3
 - dB(St(p1,p1))
 - dB(St(p2,p1))
- Port Field Display
- Field Overlays
- Radiation
- Definitions

Properties

Name	Value	Unit	Evaluated Value
Name	defect		
Material	"air"		"air"
Solve Inside	☒		
Orientation	Global		
Model	☒		
Group	Model		
Display Wi...	☐		
Material A...	☐		
Color			
Transparent	0		

Solids

- air
 - defect
 - CreateRectangle
 - CoverLines
 - WrapSheet
 - ThickenSheet
 - CloneTo
- copper
 - via
 - CreateCylinder
 - CloneTo
 - CloneTo
 - CloneTo
 - CloneTo
- silicon
 - dep
 - CreateCylinder
 - Subtract
 - CloneTo
 - sub
 - CreateBox
 - Subtract
 - silicon_dioxide
 - Subtract
 - oxide
 - CreateCylinder
 - Subtract
 - CloneTo
 - CloneTo
 - Subtract

Coordinate Systems

Planes

Named Selections

Message Manager

- *Global - Messages
- defected_cu_21_06 (C:/Users/USER/Desktop/oip_21_06/)
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 - Renormalizing the port impedance will not impact the field results if the 'Include Port Post Processing Effects' is disabled in Edit Sources. (11:28:14 AM Jun 21, 2026)
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 - Interpolating frequency sweep complete. Converged. (11:30:38 AM Jun 21, 2026)

Select Definition

Materials | Material Filters

Search Parameters

Search by Name

Search Criteria

- ☒ by Name
- ☐ by Property

Libraries

☒ Show Project definitions

☐ Select all libraries

[sys] Materials

copper [project]

copper [sys:Materials]

Polyflon Copper-clad ULTEM (tm) (keyword:POLYFLON COPPER-CLAD ULTEM (TM)) [sys:Materials]

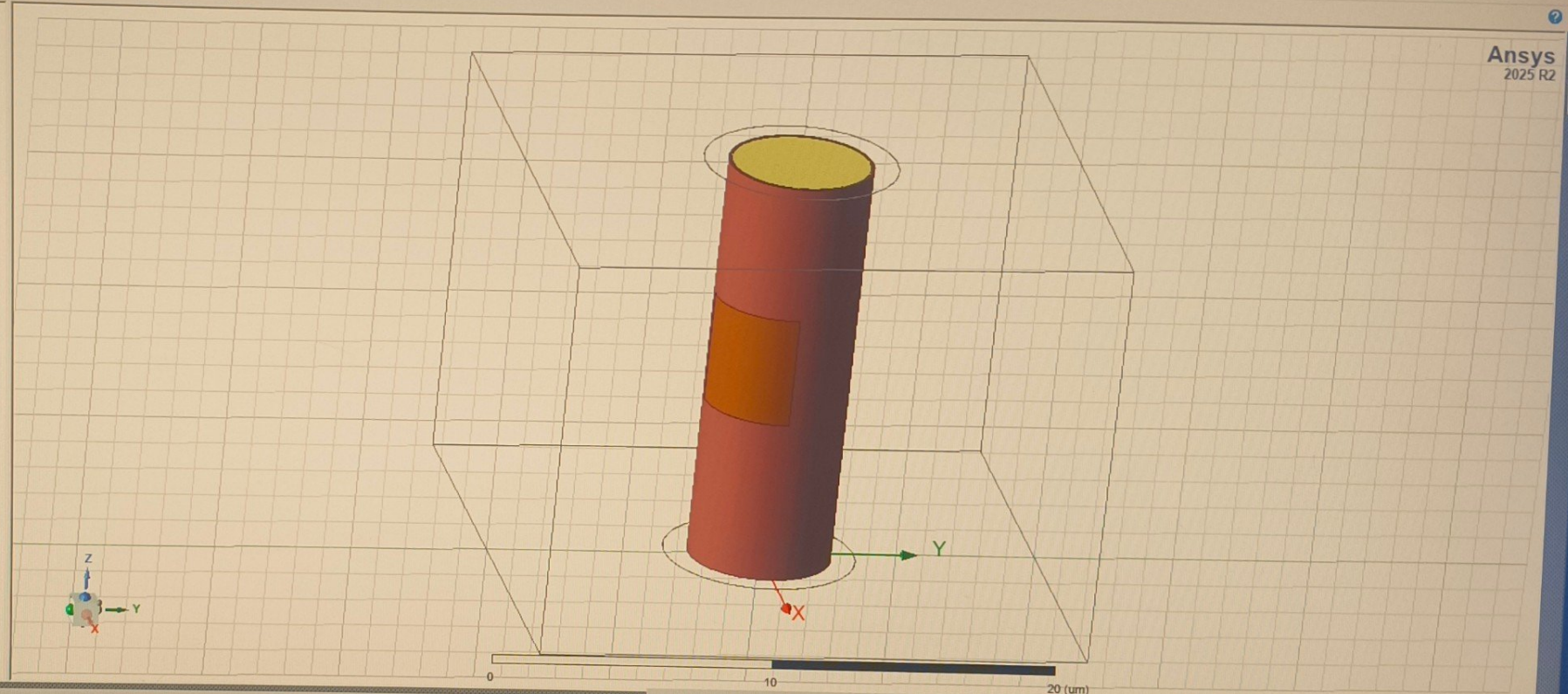
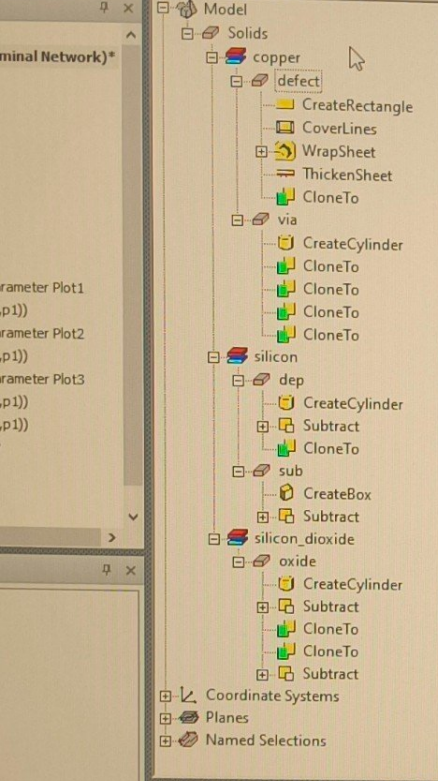
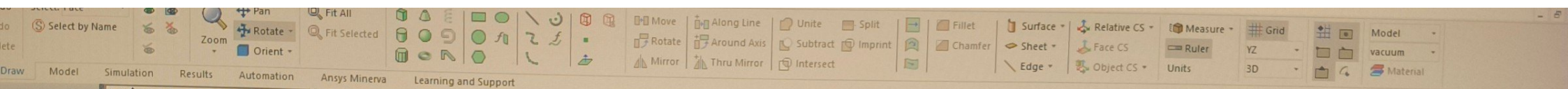
Name	Location	Origin	relative Permittivity	relative Permeability	Bulk Conductivity
copper	SysLibrary	Materials	1	0.999991	58000000
coming_glass	SysLibrary	Materials	5.75	1	0
cyanate_ester	SysLibrary	Materials	3.8	1	0
diamond	SysLibrary	Materials	16.5	1	0
diamond_hi_pres	SysLibrary	Materials	5.7	1	0
diamond_pl_cvd	SysLibrary	Materials	3.5	1	0
Dupont Type 100 HN Film (tm)	SysLibrary	Materials	3.5	1	0
Duroid (tm)	SysLibrary	Materials	2.2	1	0
epoxy_Kevlar_xy	SysLibrary	Materials	3.6	1	0
femte	SysLibrary	Materials	12	1000	0.01
FR4_epoxy	SysLibrary	Materials	4.4	1	0
gallium_arsenide	SysLibrary	Materials	12.9	1	0
IGETEK ML200/RG200 (tm)	SysLibrary	Materials	3.9	1	0

View/Edit Materials... Add Material... Clone Material(s) Remove Material(s) Export to Library...

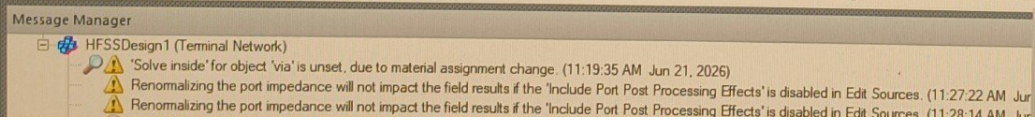
OK Cancel

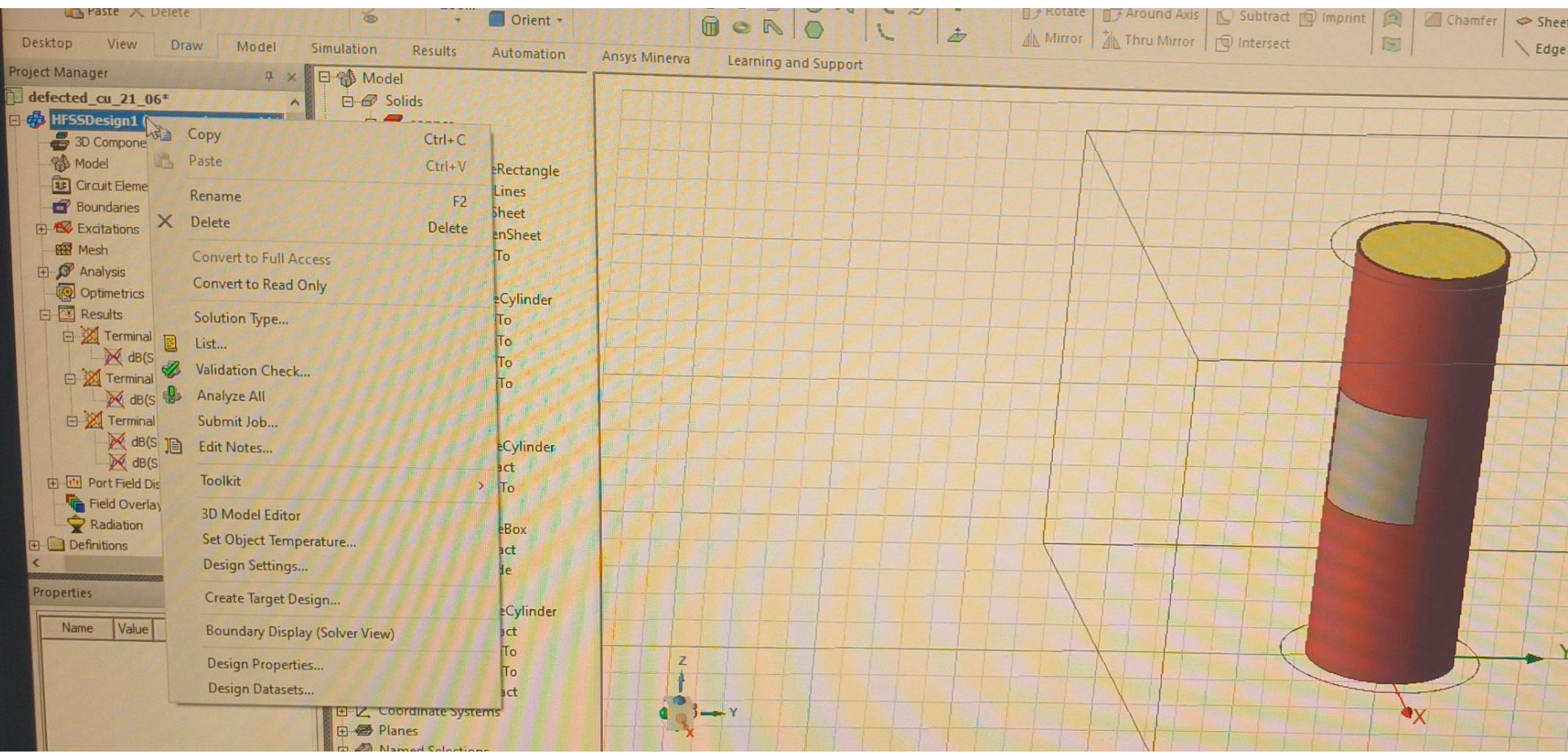
Show Hidden

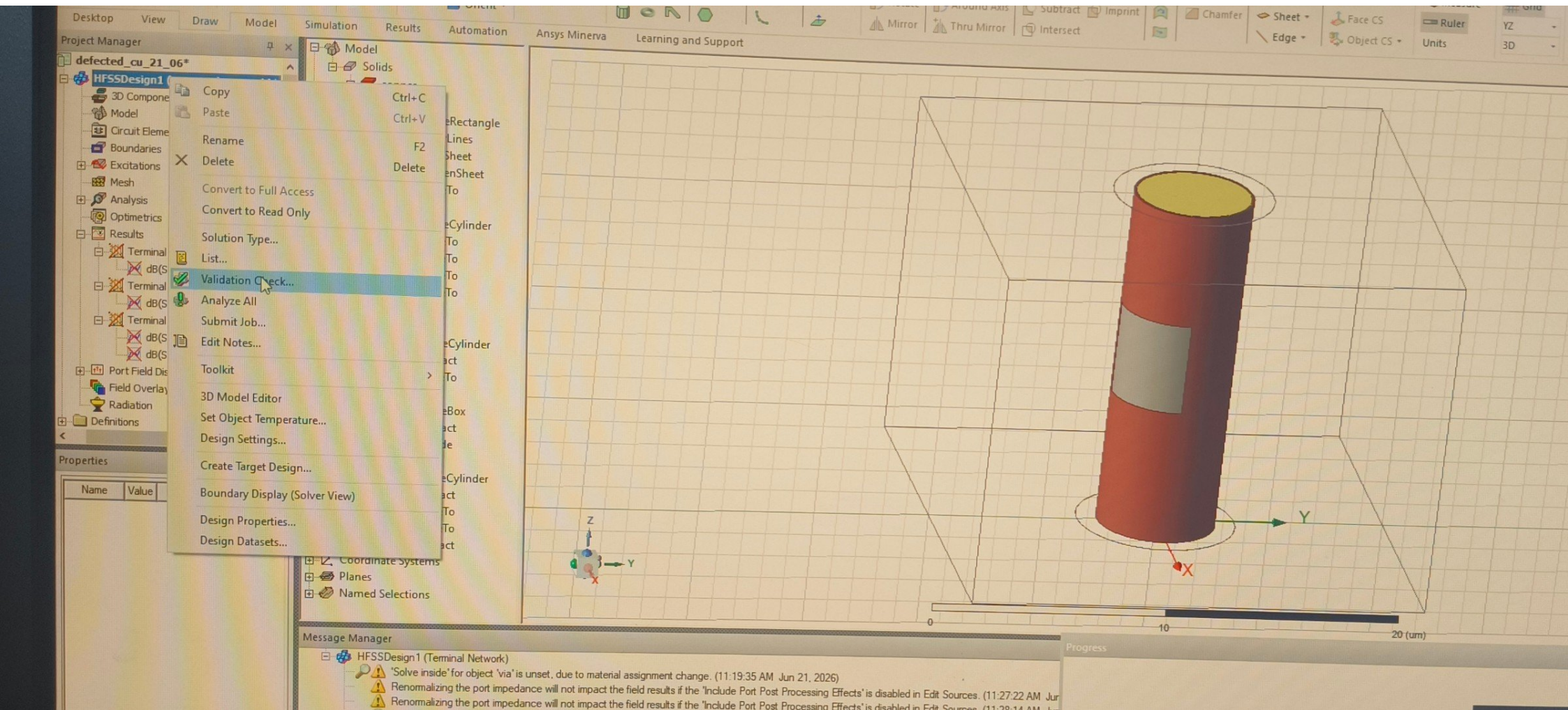
Apply

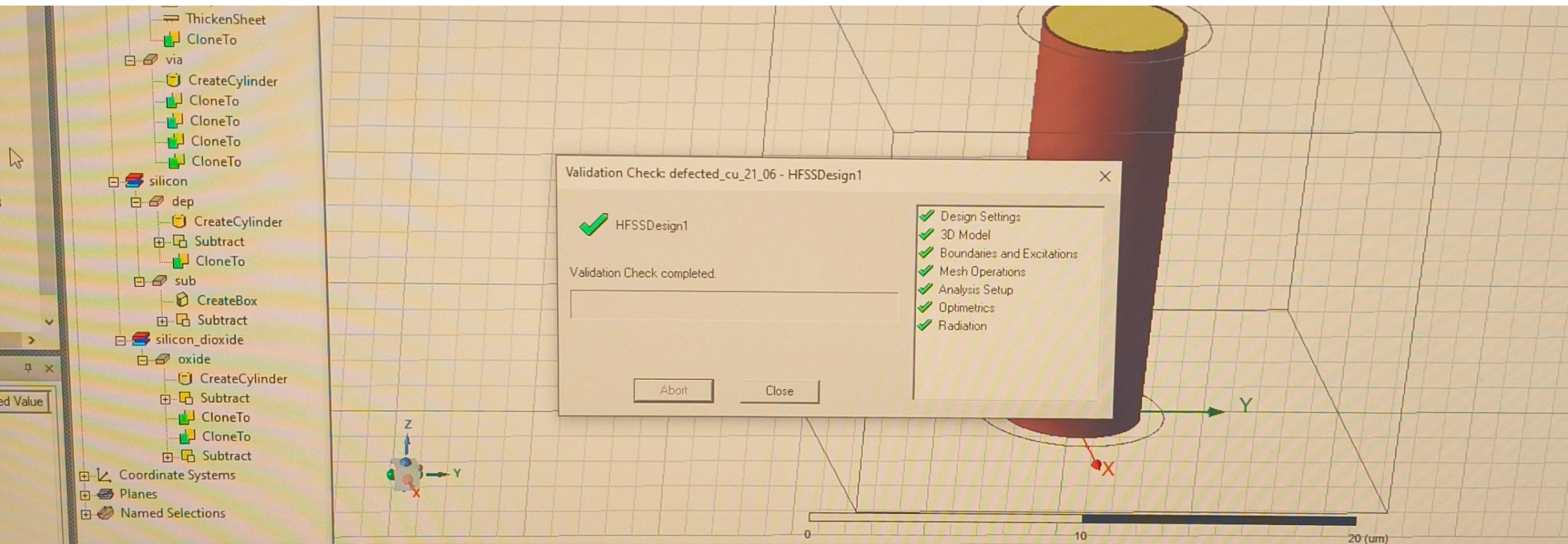


Ansys
2025 R2









Message Manager

HFSSDesign1 (Terminal Network)

- ⚠ 'Solve inside' for object 'via' is unset, due to material assignment change. (11:19:35 AM Jun 21, 2026)
- ⚠ Renormalizing the port impedance will not impact the field results if the 'Include Port Post Processing Effects' is disabled in Edit Sources. (11:27:22 AM Jun 21, 2026)
- ⚠ Renormalizing the port impedance will not impact the field results if the 'Include Port Post Processing Effects' is disabled in Edit Sources. (11:28:14 AM Jun 21, 2026)
- i An interpolating frequency sweep with 100 points has been started using HFSS - Solving Distributed. (11:30:38 AM Jun 21, 2026)
- i Interpolating frequency sweep complete. Converged. (11:30:38 AM Jun 21, 2026)
- i Normal completion of simulation on server: Local Machine. (11:30:38 AM Jun 21, 2026)
- ⚠ HFSSDesign1: Solutions have been invalidated. Undo to recover. (11:34:19 AM Jun 21, 2026)

Progress

Google Chrome

How
an FP

Ansys Electronics Desktop 2025 R2 - defected_cu_21_06 - HFSSDesign1 - 3D Modeler - [defected_cu_21_06 - HFSSDesign1 - Modeler]

File Edit View Project Draw Modeler HFSS Tools Window Help

Save Cut Copy Paste Undo Redo Select: Face Select by Name

Desktop View Draw Model Sim

Project Manager

defected_cu_21_06*

- HFSSDesign1 (Terminal Network)*
 - 3D Components
 - Model
 - Circuit Elements
 - Boundaries
 - Excitations
 - Mesh
 - Analysis
 - Optimetrics
 - Results
 - Terminal S Parameter Plot1
 - dB(St(p1,p1))
 - Terminal S Parameter Plot2
 - dB(St(p2,p1))
 - Terminal S Parameter Plot3
 - dB(St(p1,p1))
 - dB(St(p2,p1))
 - Port Field Display
 - Field Overlays
 - Radiation
 - Definitions

Properties

Name	Value	Unit	Evaluated Value
------	-------	------	-----------------

Solution Type...

List...

Validation Check...

Analyze All

Submit Job...

Edit Notes...

Toolkit

3D Model Editor

Set Object Temperature...

Design Settings...

Model

Boundaries

Excitations

Mesh

Analysis Setup

Optimetrics Analysis

Fields

Radiation

Results

Create Target Design...

Boundary Display (Solver View)

Design Properties...

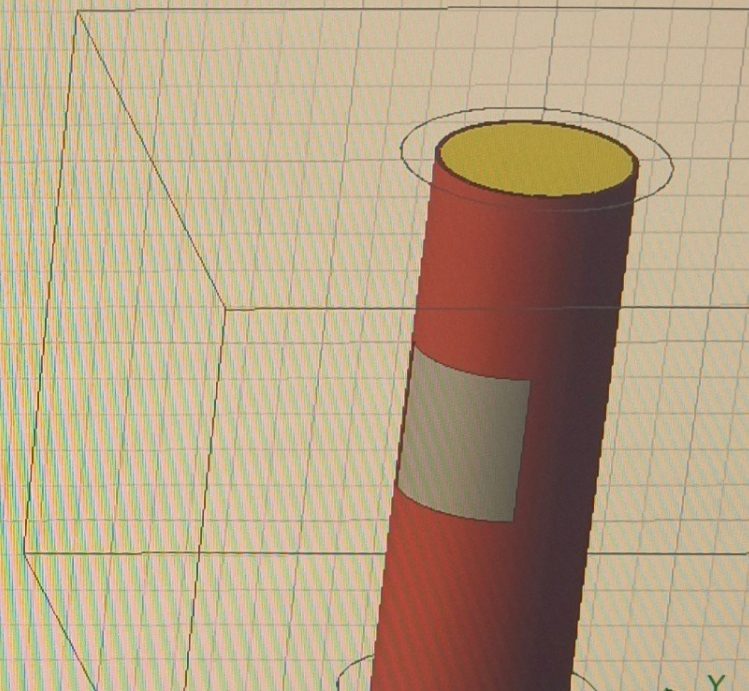
Design Datasets...

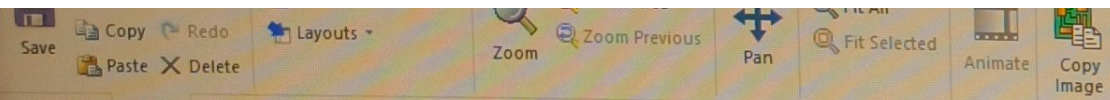
CreateCylinder

Subtract

ected Minerva Learning and Support

Move Rotate Mirror Along Line Around Axis Thru Mirror Unite Subtract Imprint Intersect Fillet Chamfer Surface Sheet Edge





Desktop View Simulation Results Automation Ansys Minerva Learning and Support

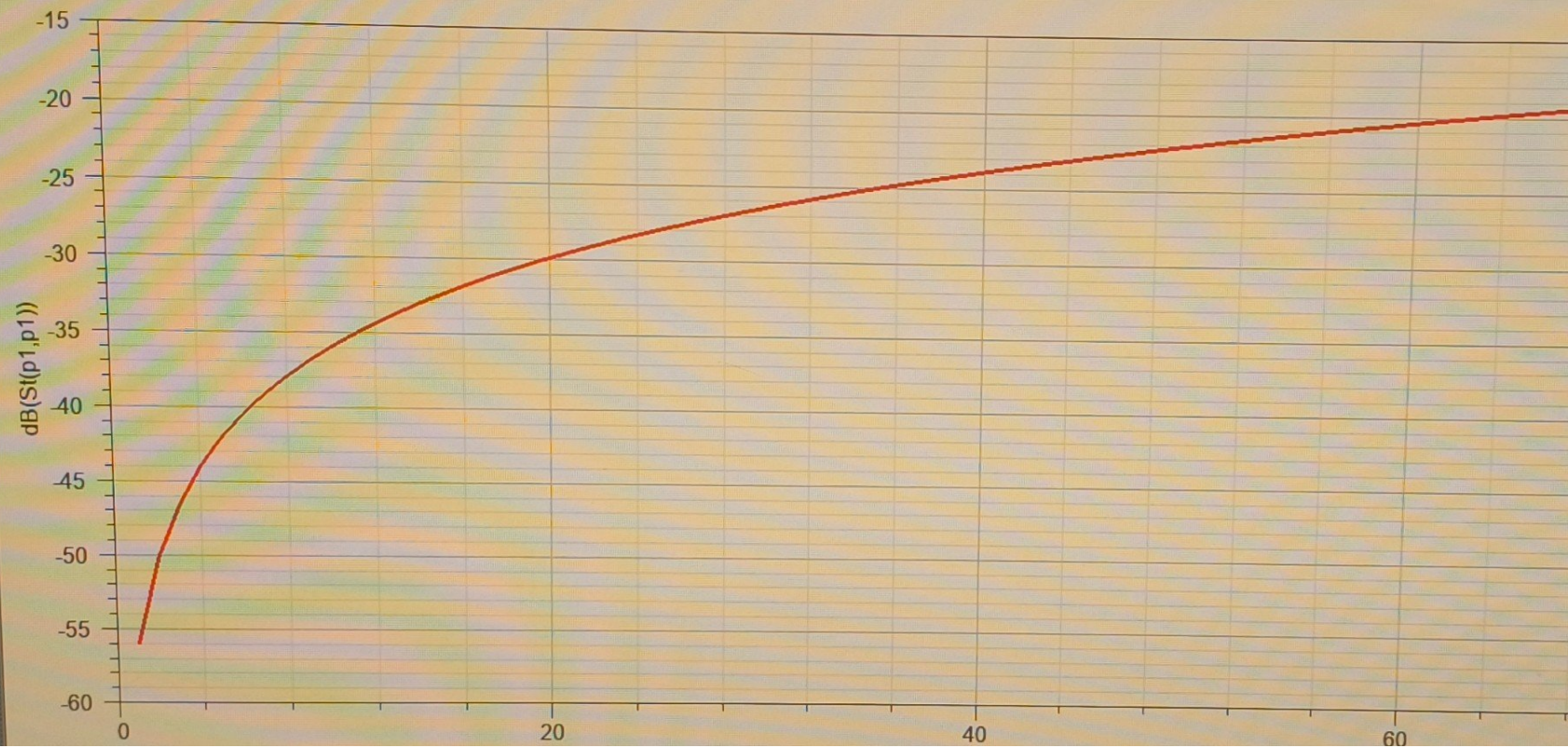
Project Manager

- defected_cu_21_06*
- HFSSDesign1 (Terminal Network)*
 - 3D Components
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 - dB(St(p2,p1))
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Properties

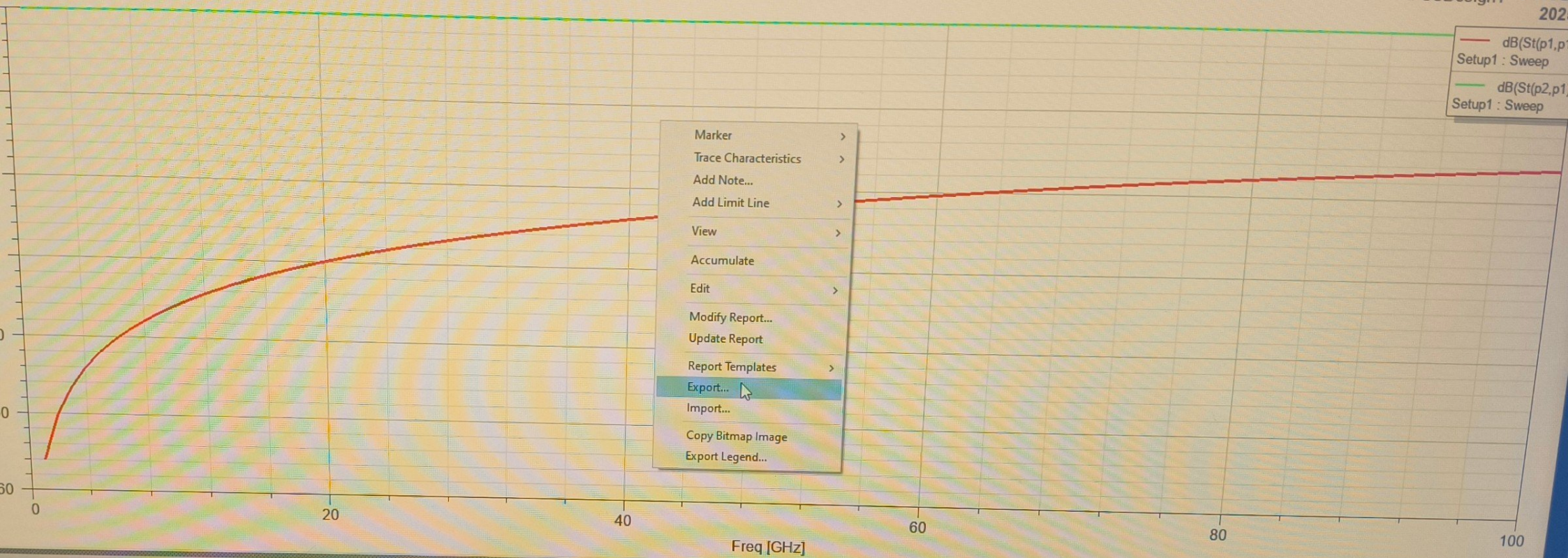
Name	Value
Name	Terminal S Parameter Plot1
Report Type	Terminal Solution Data
Display Ty...	Rectangular Plot
dB(St(p1.p...	Freq, dB(St(p1,p1))

Terminal S Parameter Plot1



Terminal S Parameter Plot3

HFSSDesign1
202



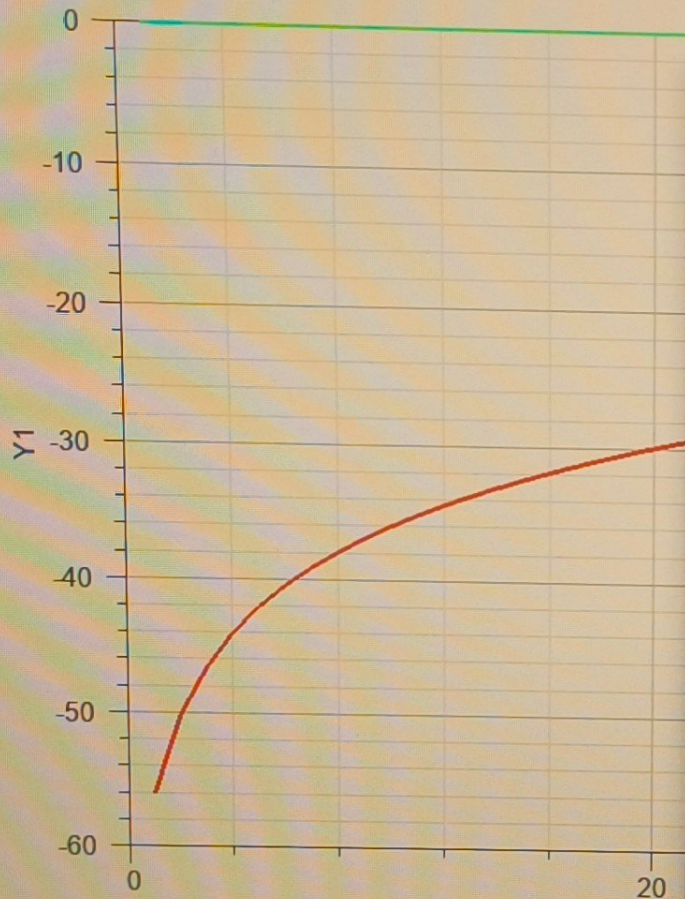
Manager

HFSSDesign1 (Terminal Network)

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Progress

Terminal S Parameter Plot3



Export Report: defected_cu_21_06 - HFSSDesign1 - Terminal S Parameter Plot3

Save in: oip_21_06

Name	Date modified	Type
defected_cu_21_06.aedtresults	21-06-2026 12:02	File f
defgectcylinder_21_06.aedtresults	21-06-2026 11:49	File f
defgectfree_21_06.aedtresults	21-06-2026 11:33	File f
spara	21-06-2026 11:42	File f

Date created: 21-06-2026 11:32
Size: 8.27 KB
Files: s_para_cylinder_air, s_para_defecfree

File name: Terminal S Parameter Plot3

Save as type: Comma delimited data files (*.csv)

Options

☐ Use Separate Columns for Curves

☐ Export Uniform Points

☒ Export with Full Sweep Range

Start: 1 GHz Stop: 100 GHz Step: 9.9 GHz

☒ Always Add Stop Value

Save Cancel